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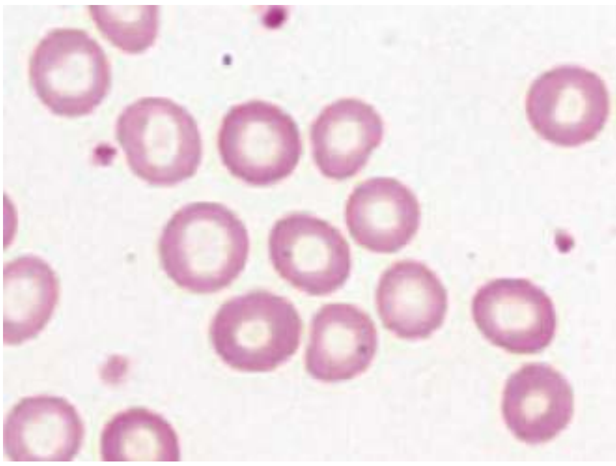
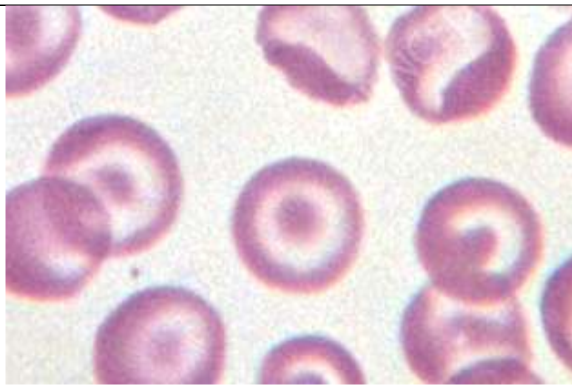
HY Heme/Onc, by Dr Michael D Mehlman

The purpose of this document is not to be a 500-page textbook with every low-yield and superfluous detail catered to. The aim is to infuse you with the highest yield info for the USMLEs.

Iron deficiency anemia vs thalassemia			
Variable	**IDA**	**Thalassemia**	**HY points**
Hb	↓	↓	- Thalassemia means abnormal synthesis of either the α or β globin chain. It presents two ways on USMLE: 1) As a microcytic anemia ***despite*** normal iron + ferritin; or 2) As a patient who has a microcytic anemia ***despite*** iron supplementation. - Most IDA is seen in menstruating women, even if their menses aren't heavy. FM forms will actually explicitly say menses are normal/not heavy, but Dx is still IDA. - FM form will give you microcytic anemia in a woman who's 32, and then the answer is just "check serum iron and ferritin." This is because thalassemia is often misdiagnosed as IDA, where if we get the iron studies back and see iron + ferritin are normal, we say, "Ok that's thalassemia, not IDA." → next best step = hemoglobin electrophoresis. - Obgyn forms like to give pregnant women at first antenatal screening with microcytic anemia who are given iron for 3 weeks but show no improvement. Q won't mention anything about patient's iron or ferritin levels. So you say, "Ok, this is a microcytic anemia *despite* iron supplementation, meaning the iron + ferritin are actually normal, so this is thalassemia." → next best step = hemoglobin electrophoresis.
MCV	↓	↓	
Serum iron	↓	Normal	
Ferritin	↓	Normal	
Hb electrophoresis	Normal	**α: normal;** **β: ↑ HbA2 + HbF** (See discussion at bottom of table)	
What happens if we give iron?	Improvement	**No improvement (HY)**	
RDW	↑ (HY)	↓/Normal (HY)	- Red cell distribution width. - All you need to know is, basically always, ↑ RDW means IDA straight-up 9/10 Qs. - The only Q where ↑ RDW is seen in thalassemia is on CMS FM form 4, where IDA is wrong, but they give ferritin within the normal range, so IDA isn't possible. But the ↑ RDW is essentially an erratum, as this is rare as fuck. - Therefore we can essentially say that ↑ RDW is sensitive but not specific for IDA, where if RDW isn't ↑, we can rule-out IDA, but be careful about automatically saying "must be IDA" if RDW is ↑.
Blood smear	Pale RBCs	Target cells	- IDA causes pale RBCs (i.e., ↑ central pallor). Not dramatic.  - Target cells are highly buzzy and pass-level for thalassemia. Probably 14/15 Qs on USMLE mentioning target cells are thalassemia (α or β, doesn't matter).

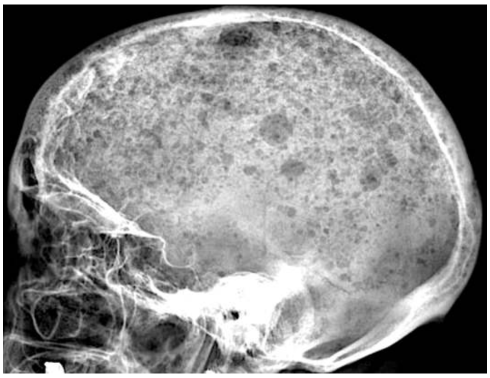
			 - There is one NBME Q with target cells in a splenectomy patient, but they're a background finding and don't facilitate answering the Q in any way.
Tx	Iron	Transfusions if severe	- Oral iron is ideal over parenteral (IV) or intramuscular iron always. I mention this because answers such as "parenteral iron" are wrong/distractors basically always. I'll say to a student, "You want to give this patient IV iron?" - Oral iron can cause constipation and black stools. USMLE wants you to know iron, aluminum (antacid), and verapamil all cause constipation. - Transfusional siderosis (secondary hemochromatosis) is sophisticated way of saying iron overload due to repeated blood transfusions, e.g., in β-thalassemia major. Each RBC transfusion contains iron. Use chelators (e.g., deferoxamine) to treat.

- Thalassemia is impaired synthesis of either the α- or β-globin chain in hemoglobin.
- α-thalassemia can have 1-4 alleles mutated; β has 2. Don't worry about obscure intermediate types, etc.
- USMLE overwhelmingly focuses on adults for both α- and β-thalassemia, since these patients are usually misdiagnosed as having iron deficiency anemia. That's why you need to know about thalassemia, so you don't misdiagnose patients with IDA.
- Adults with α-thalassemia will be asymptomatic (1 mutation), where they have an incidental microcytic anemia picked up on antenatal screening (hence Obgyn forms love thalassemia). Adults with 2 mutations have symptoms akin to IDA, but they'll fail to improve with iron supplementation. 3 α mutations (HbH disease; $\beta 4$ tetramer) will present as a sick kid; 4 α mutations (Hb Barts; $\gamma 4$ tetramer) will be fatal *in utero*.
- β-thalassemia minor will be 1 mutation in an adult who presents similar to 2 α-thalassemia mutations (i.e., similar to IDA).
- β-thalassemia major is 2 mutations and will present as a sick kid similar to 3 α-thalassemia mutations.
- It is exceedingly HY you know that β-thalassemia has $\uparrow$ HbA2 ($\alpha 2/\delta 2$) and HbF ($\alpha 2/\gamma 2$); α-thalassemia (1 or 2 mutations) has a normal hemoglobin electrophoresis.
- Sick children with thalassemia are known to get "chipmunk facies/skull" and hepatosplenomegaly occur due to $\uparrow\uparrow$ extramedullary hematopoiesis.

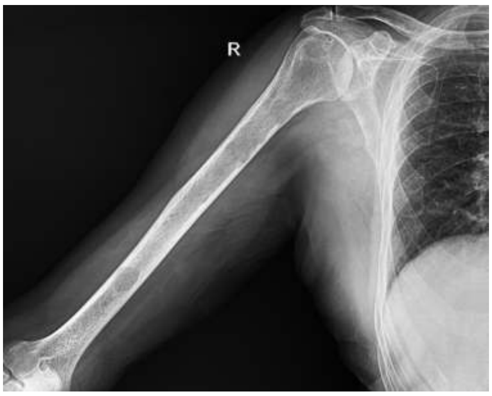
Anemia of chronic disease (AoCD)
- The answer on USMLE for anemia (↓ Hb) in patient who has **renal failure,** an autoimmune disease (i.e., RA, JRA, IBD, SLE, etc.), hepatitis B/C, HIV, or malignancy. - Patients will have low iron but **normal ferritin** (or sometimes slightly high). Ferritin is the most sensitive marker of iron stores, which means these patients are not iron deficient. - There's two main reasons iron is low in AoCD: 1) RBC production is decreased because cytokines (namely IL-6 and TNF-α) block iron release from storage sites, and 2) hepcidin, a molecule produced by the liver, decreases gut absorption of iron and promotes its storage in cells instead. - It's to my observation that **renal failure** is the highest yield cause of AoCD on USMLE. This is due to "cytokine-mediated EPO deficiency." This is an answer on an NBME form. EPO is the treatment for AoCD only if renal failure is the etiology. Otherwise the Tx is merely addressing the underlying condition (e.g., the RA), and EPO is a wrong answer. - There is an AoCD NBME Q for RA where they ask how to treat and answer is "no specific measures indicated." Student says, "Wait, I thought we were supposed to address the underlying condition, so I was confused by this answer choice." I agree. So don't take it up with me. Take it up with NBME. - One of the highest yield points I can communicate is that even though classic MCV is normal in AoCD, it will be ↓ in ~50% of NBME Qs. I've seen students say, "Oh it can't be AoCD cuz MCV is low." No. USMLE loves giving low MCV in AoCD, especially on 2CK forms. I've seen MCVs of 72 and 75 in JRA Qs.

Multiple myeloma

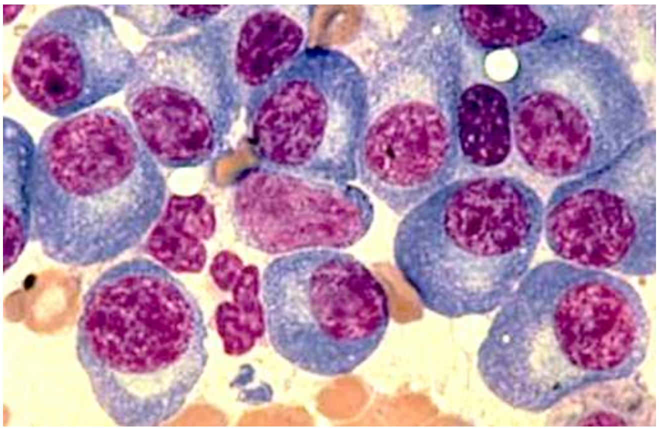
- Cancer of plasma cells.
- Plasma cells normally secrete immunoglobulins, so we have ↑↑ serum immunoglobulins.
- Multiple myeloma buzzy presentation is patient over 50 with mid-back pain + hypercalcemia.
- ↑ serum Ca^{2+} occurs due to lytic lesions of bone caused by the proliferating plasma cells. This can present as pathologic rib fractures or "pepper pot skull."



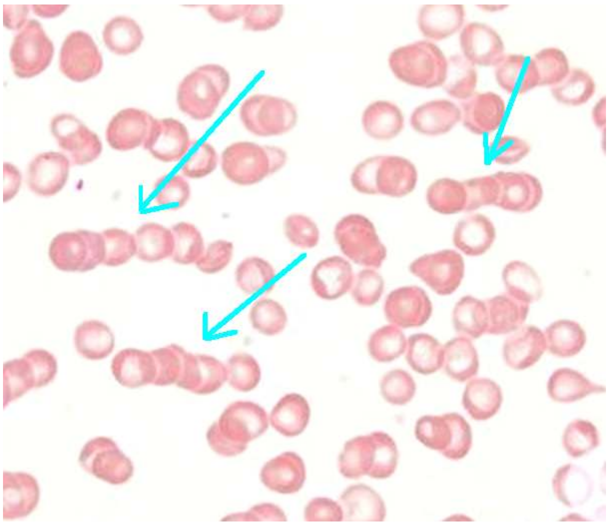
- There is NBME Q floating around where they give lytic lesions of the humerus as well; not buzzy/textbook in comparison to rib/skull lesions, but they show image similar to following:



- First step in diagnosis is **serum protein electrophoresis (SPEP),** which shows ↑↑ serum IgG kappa or lambda light chains. This is known as an M-spike (monoclonal spike; this does not mean IgM). Then **bone marrow biopsy** confirms the diagnosis, showing >10% plasma cells.
- Smear in multiple myeloma will show plasma cells (blue cells below) with "clockface chromatin," which is the appearance ascribed to the nuclei (purple below).



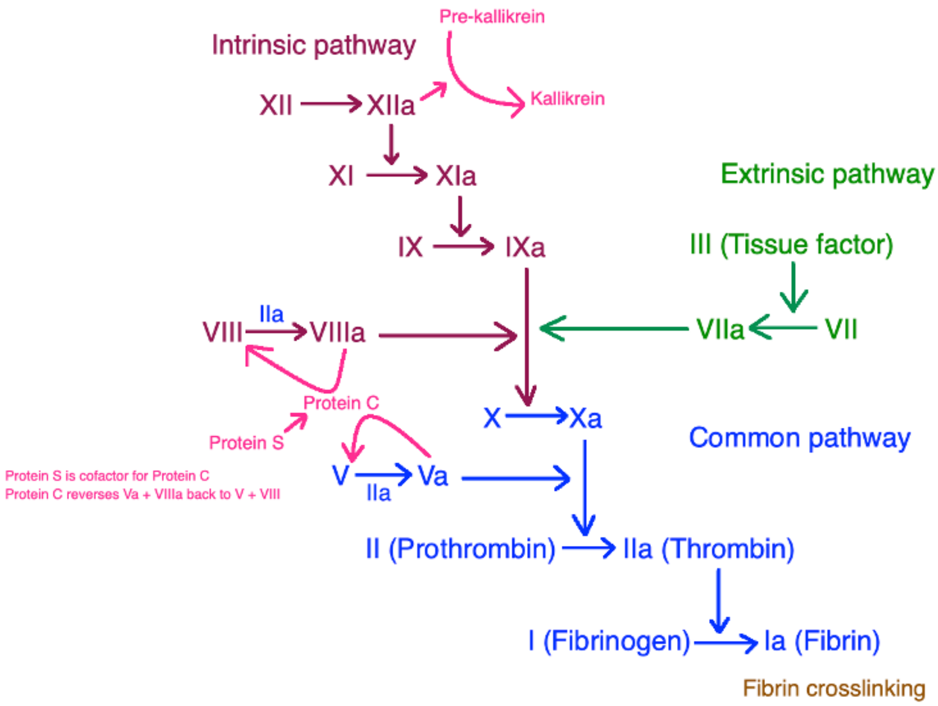
- Urinalysis will show Bence Jones proteinuria, which is simply the IgG light chains in the urine.
- Amyloidosis is proteins depositing where they shouldn't be depositing. Immunoglobulins are proteins. Since these are flying around in the blood, they deposit in the heart (**cardiac amyloidosis;** S4 heart sound with diastolic dysfunction) and kidney (**renal amyloidosis;** nephrotic syndrome).
- The IgG light chains stick to RBCs, causing the RBCs to stick to each other as stacks (i.e., Rouleaux). These are heavy and sediment quickly on centrifugation, hence ↑ ESR. USMLE can show you a picture of the Rouleaux rather than the clockface chromatin, where you need to know instantly the diagnosis is MM.



- MCV can sometimes be ↑ in MM (e.g., 108; NR 80-100). Don't be confused by this. It is common. This is caused by 1) ↑ immunoglobulins being transported by RBCs (thereby ↑ RBC size); 2) ↓ RBC production within bone marrow due to infiltrating plasma cells, causing ↑ RBC size to compensate; and 3) B9 and B12 deficiencies can be caused by MM.
- Tx is initially IV bisphosphonate to ↓ risk of bone lysis.
- Should be noted that patients with bone marrow biopsy showing <10% plasma cells have a condition known as MGUS (monoclonal gammopathy of undetermined significance). There is small annual risk of progression to multiple myeloma. These patients might have fatigue and/or ↑ MCV but no hypercalcemia or renal dysfunction.

Waldenstrom macroglobulinemia
- Cancer of plasmacytoid cells. - "-Oid" means looks like but ain't. So these are cells that look like plasma cells but they ain't plasma cells. - Causes IgM M-protein spike on SPEP (MM is IgG M-protein spike). - In contrast to MM, does not cause hypercalcemia, lytic lesions, Bence-Jones proteinuria, or amyloidosis. - Waldenstrom causes **hyperviscocity syndrome,** which will present as headache, blurry vision, tinnitus, or Raynaud phenomenon.

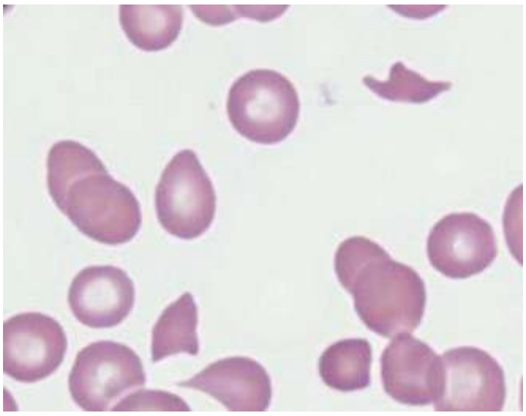
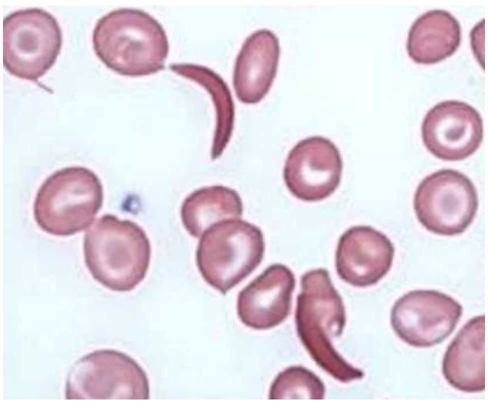
Polycythemia vera vs secondary polycythemia	
Polycythemia vera	- JAK2 mutation causing "proliferation of hematopoietic stem cells" (answer on NBME). - Qs will give ↑ Hb at 18+ (NR 12-17.5 g/dL) *and* ↑ WBCs and/or platelets. - In other words, at least 2/3 cell lines will be ↑, with RBCs always ↑. - ↑ Hb can cause hyperviscosity syndrome. - Basophilia can cause pruritis after a shower. - EPO is suppressed due to ↑ Hb. - USMLE wants **phlebotomy** as the answer to ↓ acute symptoms associated with hyperviscosity syndrome (i.e., "What is next best step to ↓ patient's headache / blurry vision?"). - Serial phlebotomy and hydroxyurea can be answers to ↓ recurrent symptoms; the USMLE will not give both at the same time.
Secondary polycythemia	- Isolated ↑ in RBCs due to ↑ EPO – i.e., WBCs and platelets are normal. - Seen in those with lung disease where arterial O2 tension is low → kidneys sense ↓ O2 and secrete EPO → acts on bone marrow to stimulate erythroid precursors. - Also seen in **renal cell carcinoma,** which can secrete EPO (and PTHrp causing hypercalcemia). - Exogenous administration (e.g., cyclists).

Brief clotting factor pathway overview


- Prothrombin time (PT) reflects the functioning of the extrinsic pathway; activated partial thromboplastin time (PTT; aPTT) reflects intrinsic pathway. - In other words, if PT alone is high, then the extrinsic pathway is fucked up. - If PTT alone is high, then the intrinsic pathway has a problem. - If both PT and PTT are elevated, then the common pathway has an issue. **- Normal PT is 10-15 seconds.** **- Normal PTT is 25-40 seconds.** **- Normal bleeding time is 2-7 *minutes*.** - Bleeding time relates to **platelets** and has nothing to do with clotting factors. - PT and PTT relate to clotting factors and have nothing to do with platelets.

HY Core DDx	
ITP	- Idiopathic/immune thrombocytopenic purpura. **- Antibodies against glycoproteins IIb/IIIa on platelets** (platelet aggregation). **- Causes ↓ platelets; ↑ bleeding time; no change PT and PTT.** - Bleeding time is high because platelets are fucked up. PT and PTT and normal because these refer to clotting factors, which have nothing to do with ITP. - Normal platelet count is 150-450,000/μL. Platelets in ITP Qs are usually <100,000. - Precipitated by viral infection, where antibodies against viral proteins cross-react with platelets (disrupts platelet aggregation); type II hypersensitivity. - Viral infection **will be asymptomatic in ~50% of vignettes.** In other words, the Q need not mention the viral infection, so don't get confused if you don't see it. - Will present two ways on USMLE: 1) School-age kid (i.e., 12) who has coryza for a few days followed by petechiae and/or nosebleeds. 2) Woman 30s-40s + no mention of viral infection + random bruising + bleeding time 9 minutes. - For the latter vignette, if they say exact same Q but bleeding time 6 minutes (normal), then answer is domestic abuse, not ITP. - There is an unusual 2CK Q on new NBME where they say girl only has heavy menses (classically a clotting factor presentation for vWD) and the answer is ITP, but vWD isn't listed and the other answer choices are clearly wrong. Just letting you know this exists. - Tx = steroids, then IVIG, then splenectomy, in that order. - If splenectomy is performed and platelets return to normal range, but then 6 months later they fall again, the answer is "accessory spleen." Sounds weird, but a small % of people have a second spleen the size of a pea that can grow in size if the main spleen is removed.
vWD	- von Willebrand disease; **autosomal dominant.** - von Willebrand factor (vWF) normally bridges glycoprotein Ib on platelets to vascular endothelium / underlying collagen (platelet adhesion). **- ↑ bleeding time; ↑ PTT (only half the time); no change PT; normal platelets.** - ↓ vWF means ↓ platelet function (and hence ↑ BT), but platelet count is unchanged. - vWF also has secondary role of stabilizing factor VIII in plasma; since we have ↓ vWF, the intrinsic pathway gets fucked up, so we have ↑ PTT. But bear in mind, this effect on factor VIII *is only a mere secondary* role, so PTT isn't always ↑. It's to my observation that only ~50% of vWD vignettes give ↑ PTT. PT is normal because the extrinsic pathway is fine. - 8/10 Qs presents as a teenager or young adult with a mix of one platelet problem and one clotting factor problem. - Platelet problem = mild, cutaneous findings (i.e., petechiae, bruising) + epistaxis. - Clotting factor problem = excessive bleeding with tooth extraction + heavy menses. (Hemarthroses are also clotting factor but are seen in hemophilia, not vWD, as I explain below).

	- For example, the Q can give 17-year-old girl who has nosebleeds + heavy periods; or 17-year-old girl who has bruising + Hx of excessive bleeding with tooth extraction (i.e., we have one platelet problem + one clotting factor problem). This is 8/10 Qs I'd say. - 1/10 Qs might give only isolated platelet or clotting factor problem (i.e., only say the patient has Hx of excessive bleeding with tooth extraction), but they'll also say *one of the parents* has Hx of nosebleeds → so you say, "Oh ok. vWD is AD, so that's your combo of platelet problem + clotting factor problem linked via autosomal dominant inheritance." - 1/10 Qs will say the patient has a cut on the finger that takes longer to heal than normal + has heavy menses + normal platelets + has normal PT and PTT + they don't mention BT. The cut on the finger is their way of saying BT is elevated. Despite only an isolated ↑ in BT here, since platelets are normal, we know it's not ITP and is instead vWD. - Ristocetin cofactor assay is abnormal. All you need to know is that this is some test that is abnormal if platelet adhesion is fucked up. Tx = desmopressin (DDAVP), which can in theory help boost functional vWD levels.
Hemophilia A+B	- Deficiency of factors VIII and IX, respectively. - **Both X-linked recessive.** - Isolated ↑ PTT; BT and PT are normal. - Factors VIII and IX are in the intrinsic pathway, so PTT is ↑. Extrinsic pathway is unaffected so PT is normal. Platelets have nothing to do with this condition, which is why bleeding time is normal. - Will present two ways on USMLE: 1) School-age boy with hemarthrosis who has isolated ↑ PTT. They can also say he has a maternal uncle who died years ago from minor head trauma. 2) Neonate who has excessive bleeding with circumcision. - Hemophilia A is treated with DDAVP and factor VIII replacement. - Hemophilia B is treated with factor IX replacement. DDAVP doesn't work apparently. - If the Q tells you a patient with hemophilia A has persistently elevated PTT despite factor VIII supplementation, the answer is "check for antibodies against factor VIII," or "check for factor VIII inhibitor" (means the same thing). Sometimes patients can develop resistance against exogenous factor replacements.
Vitamin K deficiency	- **Causes ↑ PT, ↑ PTT, no change bleeding time.** - Vitamin K is cofactor for gamma-glutamyl carboxylase, which is an enzyme that activates clotting factors II, VII, IX, and X, as well as anti-clotting proteins C and S. - Bleeding time is normal because platelets aren't affected. - Will present three ways on USMLE: 1) Neonate who has bleeding from umbilical stump +/- retinal hemorrhages. The latter can sound like shaken-baby syndrome, but the bleeding from the umbilical stump is highly buzzy for vitamin K deficiency. 2) Adult who's been on broad-spectrum antibiotics for 6 weeks for, e.g., endocarditis, who now requires a lower dose of warfarin → answer = "depletion of bowel normal flora" (Vitamin K is normally synthesized by colonic normal flora). Warfarin inhibits vitamin K function, so if Vit K is ↓, we don't need as much warfarin. 3) Exposure to rat poison, which has a warfarin-like effect (i.e., inhibits vitamin K). - Tx = supplement vitamin K, but this is slow and takes a few days to replenish clotting factors. If patient has active bleeding in the setting of vitamin K deficiency / warfarin use, fresh frozen plasma is the answer.
DIC	- Disseminated intravascular coagulation. - Runaway effect of clotting factor/platelet consumption, precipitated by multifarious etiologies such as trauma, sepsis, amniotic fluid embolism, and treatment of AML (release of Auer rods into circulation). **- ↑ BT, ↑ PT, ↑ PTT, ↑ D-dimer, ↑ plasmin activity.** **- ↓ fibrinogen, ↓ platelets, ↓ clotting factors.** - Fibrinogen is decreased because it is converted to fibrin. - Plasmin breaks down fibrin, so more fibrin means more plasmin is upregulated in an attempt to dissolve it.

	- D-dimer = fibrin degradation products; since more fibrin is being broken down, fibrin degradation products (D-dimer) ↑. - "Bleeding from catheter/IV sites" is 9 times out of 10 synonymous with DIC. However on one 2CK Surg Q, dilutional thrombocytopenia secondary to ↑ blood transfusions presents the same. - Blood smear shows **schistocytes.** 
HUS	- Hemolytic uremic syndrome. - Presents as triad of: 1) thrombocytopenia; 2) hemolytic anemia with **schistocytes;** and 3) renal insufficiency with or without hematuria. - The combination of thrombocytopenia + schistocytosis = microangiopathic hemolytic anemia (MAHA). - Mechanism is: E. coli (EHEC O157:H7) and Shigella both secrete toxins (Shiga-like toxin and Shiga toxin, respectively) that cause inflammation and damage of renal microvasculature → platelets adhere to plug damaged endothelium → platelet aggregations protrude into vascular lumen causing shearing of RBCs flying past → fragmentation of RBCs (schistocytes, aka helmet cells).
TTP	- Thrombotic thrombocytopenic purpura. - Presents as HUS triad + fever and neurologic signs. In other words: - Presents as pentad of 1) thrombocytopenia; 2) **schistocytes;** 3) renal insufficiency; 4) fever; 5) neurologic signs. - TTP is caused by antibodies against, or a mutation in, a protein called ADAMTS13, which is a metalloproteinase that breaks down vWF multimers → unable to break down platelet aggregates that plug renal microvasculature → platelet aggregations protrude into vascular lumen causing shearing of RBCs flying past → fragmentation of RBCs (schistocytes, aka helmet cells). - Can present as a stroke-like presentation in young woman, where they just want "schistocytes" as the answer (Step 1 NBME). - 2CK NBME has "plasmapheresis" as Tx.
Sickle cell	

	- **Autosomal recessive;** glutamic acid → valine missense mutation on the beta-chain. - Carrier status (one mutation) is referred to as sickle cell trait, which is less severe than sickle cell anemia (two mutations). - Step 1 NBME wants "heterozygote advantage" as the basis for the existence of sickle cell – i.e., confers resistance to malaria. - Sickle cell crises can present as **abdominal or chest pain,** as well as **dactylitis** (inflammation of the fingers). - The abdominal pain, in particular, as the initial presentation for sickle crisis has become HY on NBME exams. A new 2CK form gives RLQ pain + low Hb in child from Libya, where "blood smear" is the answer for what will confirm the diagnosis. The Q is tricky because it almost sounds like appendicitis. - Diagnosed with blood smear, followed by hemoglobin electrophoresis to detect HbS. - Offline NBME Q asks which of the following best describes the molecular basis for sickling in a patient → answer = "gain of stabilizing hydrophobic interactions in the deoxygenated form of hemoglobin S." This is because valine is more hydrophobic than glutamic acid, so we have hydrophobic interactions enabling sickling. - Another NBME has "beta chain slips into a complimentary hydrophobic pocket on the alpha chain" as answer for sickling. - Sickling notably occurs with dehydration + increased acidity. This makes sense, since the surrounding environment would become less lipophilic, meaning the HbS would consolidate/collapse unto itself to minimize surrounding surface area contact. - **Hydroxyurea ↑ HbF** and can ↓ recurrence of sickle crises. Sounds weird, but the mere presence of more HbF simply means there's fractionally less HbS floating around.
Myelofibrosis	- Usually JAK2 mutation (same as polycythemia vera). - The answer on USMLE if they say "tear-drop-shaped RBCs" or "dry tap on bone marrow aspiration." - Massive splenomegaly seen in 100% of questions. - I've seen NBME also write "tear-drop shaped poikilocytes" (fancy word that means abnormal RBCs) seen as >10% on a smear. - In myelofibrosis, the proliferation of abnormal cells in the bone marrow, notably megakaryocytes and granulocytes, leads to the release of growth factors and cytokines that stimulate deposition of collagen, resulting in bone marrow fibrosis.
Essential thrombocytosis (ET)	- Aka essential thrombocythemia. - Usually JAK2 mutation (same as myelofibrosis and polycythemia vera) causing over-proliferation of platelets. - USMLE Q will tell you patient has pain in fingertips, Raynaud phenomenon, or headache (hyperviscosity symptoms) + platelet count is, e.g., 1.4 million/µL (NR 150,000-450,000/µL). - ET is a myeloproliferative neoplasm where platelets **can be >1,000,000/µL.** - The patient will present with no specific triggering event. Compare this to reactive thrombocytosis below. - Tx = aspirin + plateletpheresis (removing platelets from blood).
Reactive thrombocytosis (RT)	- Aka reactive thrombocythemia. - Triggering event (i.e., inflammation, infection, surgery) can stimulate thrombopoietin production and increased synthesis and release of platelets. - It is to my observation the USMLE loves to give reactive thrombocytosis in the setting of **infective endocarditis,** where they'll tell you platelet count is 900,000, and the student is like, "why the fuck are the platelets 900,000." It's just RT. - ET is usually women >50 and is symptomatic; RT is anyone and is usually asymptomatic. When RT occurs, it will be a super-high platelet count you incidentally notice as part of the stem, but the focus of the Q isn't the patient's symptoms related to the ↑ platelets. - For example, there is a difficult offline NBME Q where they give neutrophilic shift with platelets of 1.4 million, where it looks like it could be RT due to infection (i.e., neutrophilic shift usually indicates bacterial infection), but the answer is ET on this form, not RT, presumably because the platelet count is >1 million (rare in RT to be this high) and the patient has symptoms (pain in fingertips).

	- I haven't seen RT as an actual correct answer on an NBME form; I've only seen it baked into questions as an incidental finding where the labs show high platelets in the setting of infection, and you, as the student, need to not be thrown off by that or think it's weird. You just say, "Oh that's probably just RT because we have endocarditis here." - Tx = address underlying infection + give aspirin.
Glanzmann	- Aka Glanzmann thrombasthenia. I've seen this on NBME as just "thrombasthenia." - Deficiency of glycoproteins IIb/IIIa on platelets → defective platelet aggregation.
Bernard-Soulier	- Deficiency of glycoprotein Ib on platelets → defective platelet adhesion. - Same as with vWD, has abnormal ristocetin cofactor assay.
SLE	- Systemic lupus erythematosus; classically women 20-40s. - Most common presenting feature is arthritis (>90%). - Malar rash too easy for most Qs and only present in maybe 1/3 of Qs, although you should know this is a type III HS. - **Thrombocytopenia** is mega-HY for Step – i.e., 32-year-old woman with arthritis + low platelets = SLE till proven otherwise. - RBCs and WBCs can also be ↓, but it's the ↓ platelets that's highest yield. The cell lines are down due to antibodies. In other words, if you get a patient with SLE where all cell lines are down, this is due to "increased peripheral destruction," not "deficient bone marrow production." - Discoid lupus = skin lesions. Mucositis = mouth ulcers. - ↑ risk of malignancy (i.e., non-Hodgkin lymphoma) in patients with autoimmune diseases (not limited to SLE, but HY) → i.e., 44-year-old woman with SLE has seizure + ring-enhancing lesion seen on CT of head → answer = primary CNS lymphoma; Toxo is wrong answer. - Causes diffuse proliferative glomerulonephritis (DPGN), which is the nephritic manifestation of lupus nephritis (i.e., SLE + red urine = DPGN till proven otherwise). - SLE in pregnant women can cause congenital heart block in neonates. Sounds weird, but shows up on NBME somewhere. - Can cause recurrent miscarriage due to anti-phospholipid antibodies. The latter are known as "lupus anticoagulant." For example, 32-year-old woman + three miscarriages + SLE → answer = anti-phospholipid syndrome. - Women with anti-phospholipid syndrome due to SLE can be administered aspirin and/or heparin during pregnancy to ↓ risk of miscarriage (warfarin is teratogen). - Women with SLE can have a false-positive syphilis VDRL/RPR test. NBME likes that. - Treat standard flares with steroids (i.e., prednisone).
PNH	- Paroxysmal nocturnal hemoglobinuria. - Red urine in patient waking up in the morning. - Mechanism is **increased complement-mediated hemolysis** caused by deficiency of CD55/59 + deficiency of GPI anchor, which protect RBCs from complement-mediated breakdown. - Q can mention red urine in patient in the morning + deficiency of GPI anchor in the stem, and then the answer will just be "complement-mediated hemolysis."
Hageman factor deficiency	- Hageman factor = factor XII, the first clotting factor in the intrinsic pathway. - All you need to know is that there is such thing as "Hageman factor deficiency" that presents as a completely asymptomatic/healthy adult who incidentally is discovered to have a ↑↑ PTT. - If factor XII is low, then XIIa will also be low, which means less pre-kallikrein is converted to kallikrein. - There is one retired Step 1 NBME Q where the answer is "deficiency of kallikrein" in the setting of an asymptomatic 60-year-old dude who has mega ↑↑ PTT. - You can call it dumb all you want. We're in agreement here. Doesn't change the fact it's on the NBME. Students have probably encountered the question not even realizing this is what they saw, hence first time you're hearing about this.

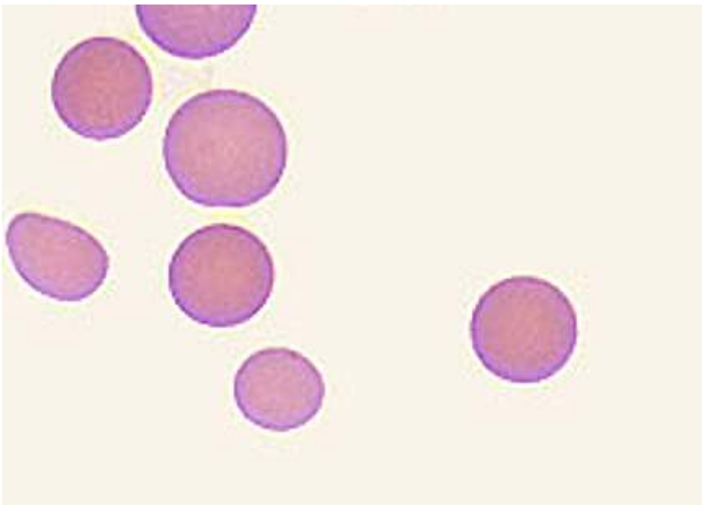
Thrombotic disorders	
Factor V Leiden	- Just be aware that there is a condition called FVL, which means the patient will have ↑ clots / DVTs. - Normally factor Va and VIIIa are reversed to V and VIII by protein C (protein S is a cofactor for protein C). In FVL, we have "activated protein C resistance," which means Factor Va is resistant to cleavage by protein C. - Conditions such as Prothrombin mutation and MTHFR also cause ↑ clots, but are lower yield.
Protein C deficiency	- Causes clots/DVT, same as FVL, but is notably associated with skin necrosis if the patient receives warfarin.
Antithrombin III deficiency	- AT III is required to reverse IIa (thrombin), Xa, and IXa back to inactive II, X, IX. - We normally think about AT III being upregulated by heparin (anti-coagulation). But patients can be deficient in AT III, causing hypercoagulable state. - Presents two ways on USMLE: 1) Young, otherwise healthy adult who has idiopathic thromboses + the Q mentions nothing about PT or PTT. 2) Patient with nephrotic syndrome who has DVT, superficial thrombophlebitis, renal vein thrombosis, or varicocele. AT III is a protein that can be lost in the urine. - In other words, patient who has peripheral edema + DVT, you should immediately think, "Oh that's DVT due to AT III deficiency from nephrotic syndrome."
Antiphospholipid syndrome (APS)	- The answer on USMLE in 3 situations: 1) Random patient with thromboses despite ↑ PTT. 2) Woman with who has recurrent miscarriages, especially if she has SLE. 3) Patient with false(+) syphilis VDRL/RPR testing. - Antibodies against phospholipids cause *in vivo* thromboses despite a paradoxical *in vitro* ↑ PTT (i.e., if PTT is ↑, you'd think we'd have bleeding diathesis, not thromboses). - PTT is ↑ because we need phospholipids for the *in vitro* assay to work. - *In vivo*, we have thromboses however because the antibodies bind to vascular walls causing endothelial cell dysfunction and prothrombotic state. - Can be idiopathic (i.e., caused by antibodies against β2-microglobulin or cardiolipin), or **secondary to SLE.** - If APS is due to SLE, we merely call these antibodies "lupus anticoagulant," but they're the same antibodies. - APS can cause recurrent miscarriages in pregnant women due to clots within the microvasculature supplying the placenta. On 2CK Obgyn forms, the answer can show up as "uteroplacental insufficiency" in, e.g.., a patient with SLE who has recurrent miscarriages. - For whatever magical reason, APS can cause **false(+) VDRL/RPR syphilis** tests.

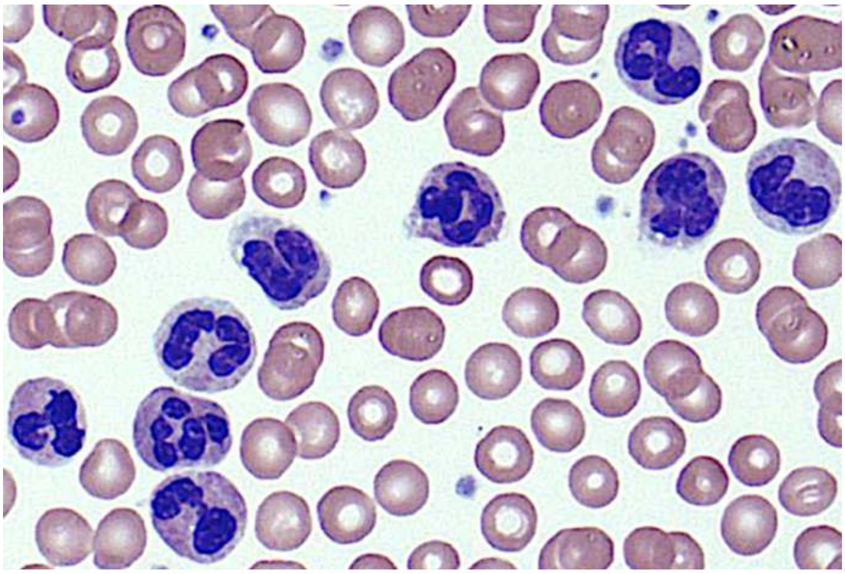
HY Neutropenia conditions	
- A HY general point for USMLE is that neutropenia can present as **mouth ulcers, fever, and sore throat.** - Questions will not usually come out and say, "Yes, the kid has neutropenia." The situation might be, e.g., a kid who's undergoing chemotherapy for AML who now has mouth ulcers and/or sore throat. - The mouth ulcers can also be described as "ulcerative pharyngitis."	
Neutropenic fever (Febrile neutropenia)	- Neutropenia + fever, LOL! - Can occur in many HY settings on USMLE: - Chemo- or radiotherapy-induced pancytopenia (i.e., RBCs, WBCs, and platelets are all ↓). - Viral-induced neutropenia (i.e., Parvovirus B19), where you see that the patient has fever + neutrophils are low (+/- RBCs and platelets). - Drug-induced neutropenia – i.e., clozapine, methotrexate, PTU, methimazole, and TMP/SMX.

	- Medical emergency (i.e., the patient has infection but no ability to fight it). USMLE wants "IV broad-spectrum antibiotics" as the next best step. - After Abx, the USMLE wants granulocyte colony-stimulating factor (i.e., G-CSF, or filgrastim) as the answer. - I've never seen granulocyte transfusion as a correct answer on USMLE.
Chemo-/radiotherapy-induced neutropenia	- USMLE might give you patient who's receiving chemo and/or radiotherapy who has pancytopenia + fever, where they want antibiotics as the answer. - It's to my observation students often erroneously choose "give RBCs" or "give platelets" in this setting, where they'll say, "but the Hb is <7g/dL." But USMLE wants "broad-spectrum antibiotics" for neutropenic fever as first step, even when the patient has pancytopenia. If the patient is afebrile, we would probably address ABCs first.
Viral-induced neutropenia	- Parvovirus B19 can infect myeloid precursors and cause neutropenia – i.e., viral-induced neutropenia. - Other viruses can cause formation of antibodies against neutrophils, similar to how they can cause antibodies against platelets in ITP.
Drug-induced neutropenia	- Clozapine, methotrexate, propylthiouracil, methimazole and TMP/SMX are the highest yield drugs that cause neutropenia on USMLE. There are others, but LY. - If Q says patient being treated for schizophrenia develops mouth ulcers → answer = neutropenia due to clozapine. - If Q says patient was recently treated in hospital for hyperthyroidism + now has ulcerative pharyngitis → answer = "drug-induced neutropenia," probably due to propylthiouracil, which is often used for thyrotoxicosis.
Cyclic neutropenia	- Gene mutation causing episodes of fever + mouth ulcers in a kid age 1-3. These episodes typically occur every 21 days, but the Q might not specify this number and will just say, "There have been several episodes like this before."

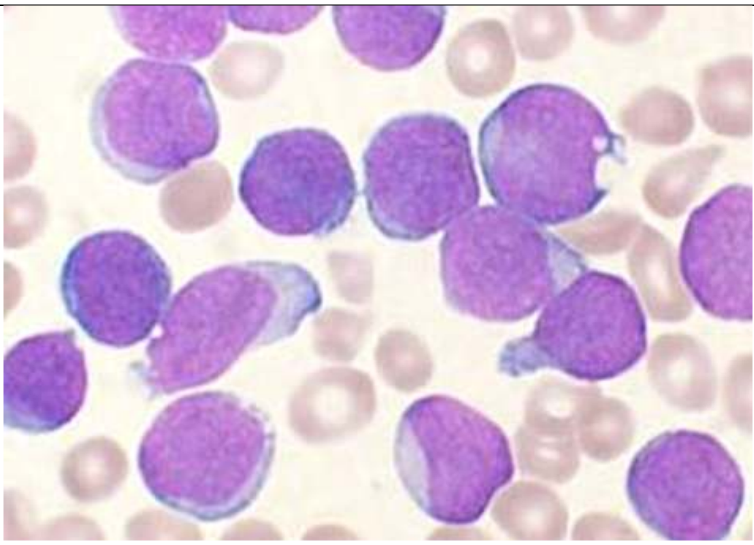
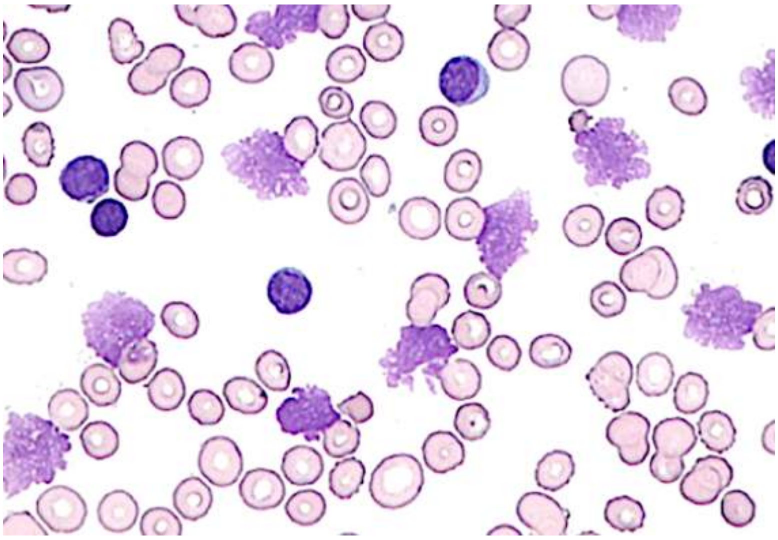
Other anemias	
Aplastic anemia	- Means all cell lines (i.e., RBCs, WBCs, platelets) are ↓ due to ↓ bone marrow production. - Can be due to Parvo B19, where it infects not just myeloid progenitors, causing neutropenia, but also erythroid progenitors and megakaryocytes, causing ↓ RBCs and platelets, respectively. - Can also be due to other viruses, such as HepA or HIV. - Can be chemo- and/or radiotherapy-induced. - Fanconi anemia is rare, AR aplastic anemia that, for whatever reason, also presents with hypo-/aplastic thumbs/radii.
Pure-RBC aplasia	- Means only RBCs are ↓ (i.e., WBCs and platelets are normal). - Can be caused by Parvo B19. - Thymoma can cause pure-RBC aplasia (as well as myasthenia gravis). - Diamond-Blackfan anemia is rare pure-RBC aplasia that presents with triphalangeal thumbs.

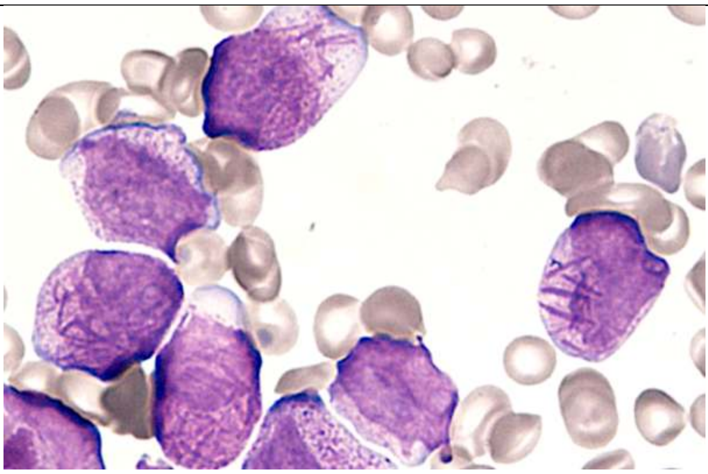
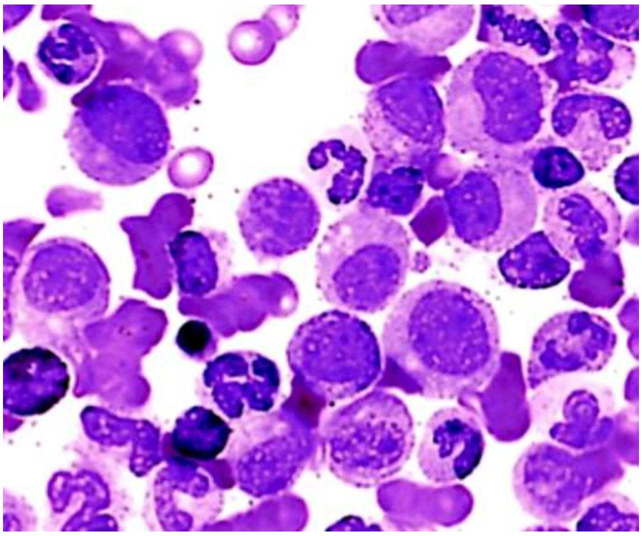
Spherocytosis	
Hereditary	- **Autosomal dominant** (one of the highest yield inheritance patterns on USMLE). - Heterozygous mutation in ankyrin, spectrin, or band proteins, causing an RBC cell membrane/cytoskeletal defect. This causes the RBC to deform away from the normal biconcave disc and into a spherical shape. - The USMLE might give you hereditary spherocytosis and then the answer is just "cell membrane," or "cytoskeleton," or "cytoskeletal defect."

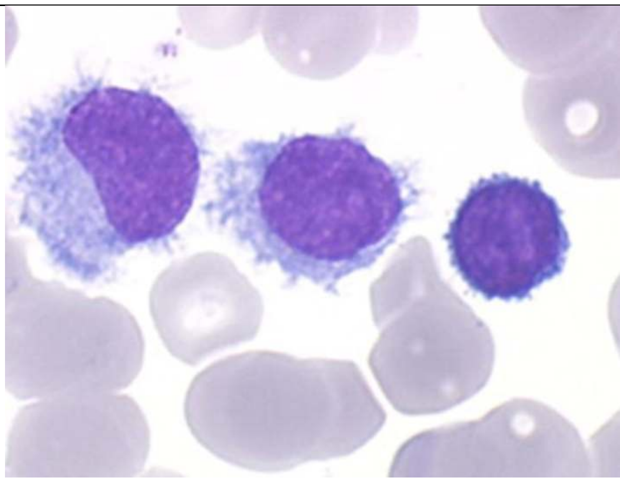
	- Can be described as RBCs lacking central pallor. If they show you a smear, they look full and red.  - Causes a normochromic, normocytic anemia (asked on an NBME directly). - Can cause **pigment stone cholelithiasis** due to the spleen identifying the spherocytes as abnormal and increasing RBC turnover. - RBC mean corpuscular hemoglobin concentration (MCHC) can be ↑. This is not specific to spherocytosis but is nevertheless a rare lab parameter to see in Qs. It's to my observation that if they mention it, 4/5 times the Dx is spherocytosis. - Diagnose via osmotic fragility test and eosin-5-maleimide. - Coombs test is negative (meaning we don't have antibodies against RBCs). This is important in terms of contrasting with spherocytosis due to hemolysis (as I discuss below). - Treatment is splenectomy (to ↓ RBC turnover). The answer can appear on USMLE as "splenectomy and cholecystectomy" together. - USMLE can mention a child who has a low Hb + one of the parents had splenectomy +/- cholecystectomy performed in the past for "an anemia" → answer = hereditary spherocytosis. I also point out to the student the AD inheritance pattern based on one of the parents having the condition as well. - Q can also just ask straight-up which heme condition is often treated with splenectomy, and the answer is hereditary spherocytosis; students sometimes erroneously choose sickle cell. Remember, sickle cell causes autosplenectomy due to repeated microinfarcts within splenic microvasculature; it is not treated with splenectomy.
Hemolytic anemia	- Spherocytosis is not always due to hereditary spherocytosis and can sometimes be seen with drug- or infection-induced hemolysis. - In this case, the Coombs test is (+), meaning we have antibodies against RBCs. In contrast, with hereditary spherocytosis, the mechanism has nothing to do with antibodies, so Coombs test is (-). - Antibodies binding to RBCs can trigger complement-mediated damage, which can lead to deformation of the RBC membrane and cytoskeleton. - USMLE will give kid with recent viral infection who has low Hb and spherocytes on a smear + positive Coombs; answer = hemolytic anemia, not hereditary spherocytosis (i.e., answer = "no specific mutation").

Leukemoid reaction
- Exaggerated immune response to infection in which WBCs go ↑↑ and can hence resemble leukemia. - As I mentioned before, "-oid" means "looks like but ain't," so the patient's labs can look like leukemia (because the WBCs are so high), but it's not leukemia. - Normal leukocyte count is 4-11,000/μL. Typical infections might cause WBCs to go ↑ to the teens or 20,000s/μL. But when WBC levels creep into the 30s, this becomes increasingly rare, even for patients who are septic, so we want to start thinking about leukemia. This is why if the patient doesn't turn out to have leukemia we call it leukemoid reaction. - This will be the answer on USMLE for patient who has **infection + WBCs >30,000/μL, PLUS** they give you a blood smear that shows neutrophils or they say leukocyte ALP is elevated (in CML it's low).  - This is the smear the USMLE will show you if they want leukemoid reaction. This is showing you **neutrophilia** (i.e., just lots of neutrophils). They can also write the answer as **reactive granulocytosis,** or as "increased release of leukocytes from bone marrow post-mitotic reserve pool." - A typical leukemoid reaction Q might give you a UTI + WBCs 32,000/μL + the above smear. Not hard.

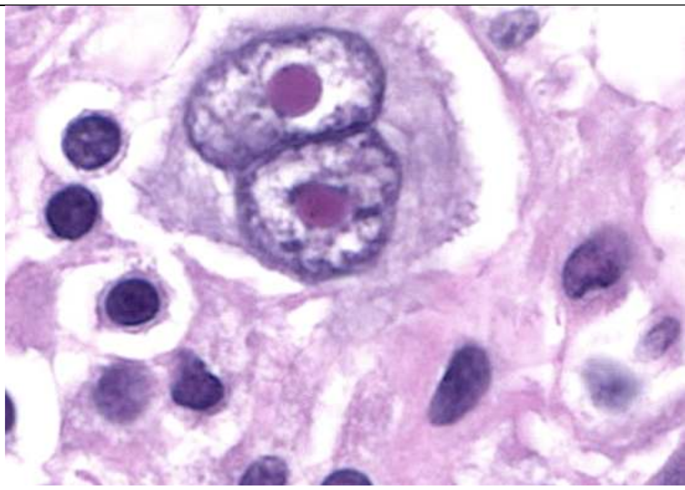
HY Leukemias	
- A leukemia is a cancer of white blood cells in the blood. - They are almost always B-cell in origin. - Apart from Sezary syndrome and mycosis fungoides (discussed below), if one of the HY lymphocytic leukemias (i.e., ALL, CLL) is T-cell in origin, USMLE will give an SVC-like syndrome (i.e., positive Pemberton sign with congested neck veins and/or facial erythema/swelling) due to a thymic lesion. - Patients can sometimes get bone pain with leukemia (i.e., USMLE Q can say patient has leg/shin pains).	
ALL	- Acute lymphoblastic leukemia. - The answer on USMLE for a kid with leukemia (i.e., pre-adolescent). - Q will give you kid who's, e.g., 6-years-old, and has leukocyte count of 60,000/μL, where it's 90% lymphocytes. - **An important DDx of ALL is pertussis,** that for whatever reason, can cause an ALL-like picture where WBCs can be 30-40,000+, where it's 90% lymphocytes. This is called reactive lymphocytosis. The distinction with ALL is that, in pertussis, they'll say there's a cough and/or post-tussive emesis (vomiting after cough) and/or hypoglycemia (HY finding in pertussis). - ALL can be CD10 and TdT positive. Dumb, but I've received feedback of them showing up. - ↑ Risk with Down syndrome. The Q can give you standard vignette of Downs and then ask for what child is at greatest risk of developing → answer = "excess lymphoblasts."

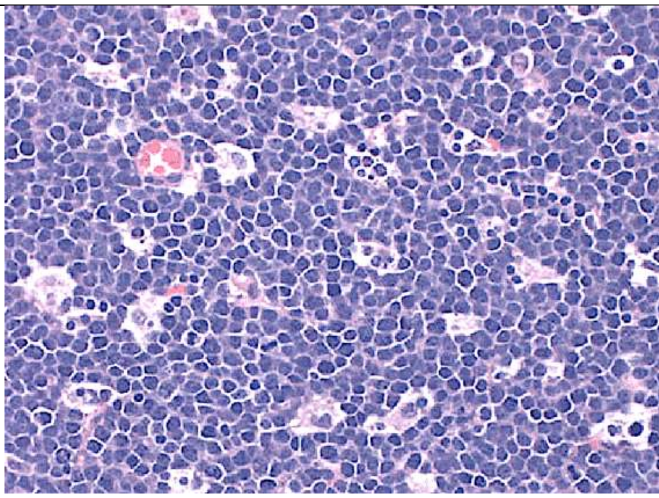
	 ALL; lymphoblasts appear smooth and relatively uniform.
CLL	- Chronic lymphocytic leukemia. - Vignette will sound just like ALL but not in a kid (usually older adults). - The Q might say 70-year-old + WBCs 87,000/µL (90% lymphocytes), where they ask for the next best step → answer = “quantitative immunoglobulin assay,” which can show ↑ monoclonal immunoglobulin sometimes in CLL. Sounds nitpicky, but it’s an answer on an NBME. - CLL smear can show smudge cells, which are fragile cells on LM.  - Leukemic cells are CD5 and CD23 positive (asked on an NBME). - Can be associated with warm autoimmune hemolytic anemia (IgG against RBCs).
AML	- Acute myeloblastic leukemia. - Will show Auer rods on a smear.

<table>
<tr><td></td><td>
- The image of Auer rods is exceedingly HY for the Step. They are composed of myeloperoxidase, which is a blue-green heme-containing pigment.
- If an AML Q doesn't give you the buzzy image of Auer rods, they'll say there's "30% blasts" or "50% blasts."
- Acute promyelocytic leukemia (APL; aka AML type M3) has a t(15;17) translocation.
- Tx is all-trans retinoic acid (vitamin A). This causes the maturation of leukemic cells into mature granulocytes.
- Tx of AML also leads to leukemic cell lysis, where release of Auer rods into the circulation can precipitate DIC.</td></tr>
<tr><td>CML</td><td>- Chronic myelogenous leukemia.
- Caused by t(9;22) translocation (Philadelphia chromosome).
- Results in formation of bcr/abl tyrosine oncogenic tyrosine kinase, which is a "fusion protein."
- The stem will say there's all sorts of myelo-sounding cells – i.e., myelocytes, promyelocytes, metamyelocytes. 9/10 Qs that mention these cells are CML.
- Leukocyte ALP is low.
- Smear shows what I refer to as a "motley mix" or "soup," which shows all different types of cells.

- This is HY smear that differentiates it from leukemoid reaction (which instead shows neutrophilia).
- Tx is with imatinib, which targets the bcr/abl tyrosine kinase.</td></tr>
<tr><td>Hairy cell</td><td>- Q will merely say patient has leukemia characterized by "cytoplasmic projections" that stains positive for tartrate-resistant acid phosphatase (TRAP).</td></tr>
</table>

	 - The stem can give low WBCs, which is unusual for leukemia. 9 out of 10 questions for leukemia will give ultra-HY WBCs. The NBME Q I've seen on hairy cell gives image similar to above + gives WBCs 3,500 + they say WBCs with cytoplasmic projections" + stains positive for TRAP.
Sezary Syndrome	- Cutaneous T-cell lymphoma that extends to the blood as a T-cell leukemia. - Caused by human T-cell lymphotropic virus (HTLV). - Has cerebriform-shaped T cells in blood stream.  - Presents classically as erythroderma (red skin).

Lymphomas	
- A lymphoma is a cancer of the lymphatic system. - Usually starts in lymph nodes, but can also be present in other areas of the lymphatic system, such as the spleen and bone marrow. - Almost always B-cell in origin. - Lymphomas are categorized as Hodgkin or non-Hodgkin (NHL). - Epstein-Barr virus, HIV, general immunosuppression (i.e., chronic corticosteroids or immunosuppressant drugs), and autoimmune disease (e.g., SLE, Hashimoto) increase the risk for lymphomas.	
Hodgkin	- Characterized by Reed-Sternberg cells, which are CD15/30 (+) B cells with a characteristic "owl eye" appearance.

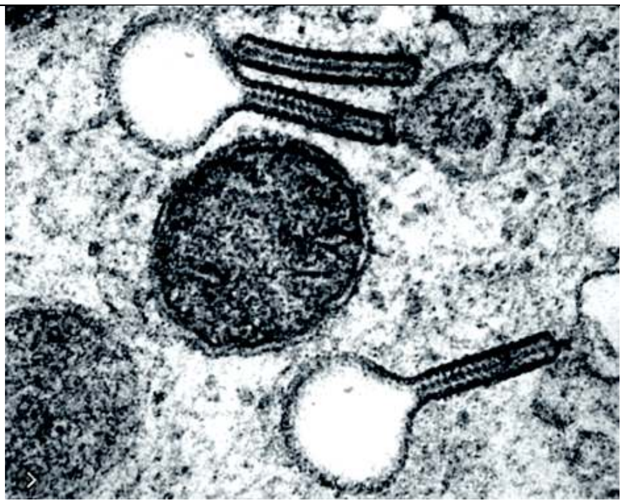
	 - 100% of questions give a painless lateral neck mass or facial swelling (lymphadenopathy). - The USMLE Q will then give at least one additional finding of: 1) hepatomegaly; 2) mediastinal mass (not thymoma; this is mediastinal lymphadenopathy); 3) Virchow node (Troisier sign of malignancy; palpable left supraclavicular lymph node); this is not limited to gastric cancers; I've seen an NBME Q where they give this latter finding for Hodgkin. - B symptoms (i.e., fever, night sweats, weight loss) are common but not mandatory in vignettes. - So for example, they'll say 21-year-old guy with 1-month history of painless left-sided neck/facial swelling + mediastinal mass. Diagnosis? $\rightarrow$ Hodgkin. Very buzzy/easy. Students somehow get flummoxed and emotional that it's not thymoma. - Question can give normal leukocyte count and differential. An NBME Q gives WBCs of 10,000/µL (NR 4-11,000) and lymphocytes of 33% (NR 25-33). This is because lymphomas are cancers of lymphatic tissue. They're not leukemias, which are cancerous WBCs floating around the blood. - Nodular-sclerosing type tends to favor women. - $\uparrow$ Normal B cells relative to abnormal RS cells = better prognosis. For example, leukocyte-rich Hodgkin has $\uparrow$ normal B cells in comparison to RS cells and has better prognosis; in contrast, leukocyte-deplete Hodgkin has comparatively more RS cells and worse prognosis.
Burkitt	- t(8;14) translocation of c-myc gene; can rarely be t(2;8) or t(8;22). - c-myc is a **transcription factor.** - Presents classically as jaw lesion in African boy, or as abdominal lesion.  - Histo is buzzy "starry sky" appearance, which is a background of basophilic (purple) B cells with translucent macrophages.

	 - Sounds odd, but NBME wants you to know the macrophages are called “tingible body” macrophages (not tangible), which phagocytose the lymphomatous B cells undergoing **apoptosis.** An NBME Q points to a tingible body macrophage and then the answer is just “apoptosis.” The macrophage itself isn’t undergoing apoptosis; it’s merely phagocytosing the apoptotic B cells characteristic of Burkitt. You might ask, “If the B cells are undergoing apoptosis, then how is the cancer growing/spreading?” The answer is that the rate of proliferation far exceeds that of the apoptosis.
Follicular	- t(14;18) translocation of *Bcl-2* gene, which codes for an **anti-apoptotic molecule.** - Most common indolent NHL (which means less aggressive). - Presents as waxing/waning painless neck mass over 1-2 years.
DLBCL	- Diffuse large B cell lymphoma. - Most common aggressive NHL.
Mantle cell	- t(11;14) translocation involving over-expression of cyclin D → can’t halt cell cycle.
Mycosis fungoides	- Cutaneous T-cell lymphoma characterized by cerebriform-shaped cells. - Less aggressive than the T-cell lymphoma of Sezary syndrome and does not extend to the blood as a T-cell leukemia.  - An important differential for “skin rash.”

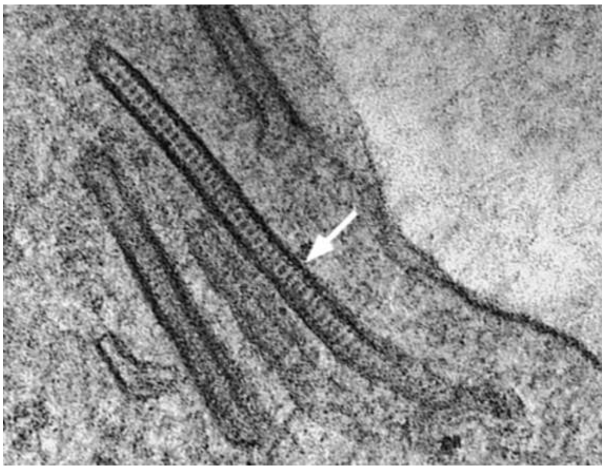
Warm vs cold autoimmune hemolytic anemia (AIHA)	
- Both present with (+) Coombs test, which just means antibodies against RBCs. - In other words, if a Q says Coombs is (+), we have Abs against RBCs; if it's (-), we don't. - USMLE doesn't give a fuck about you memorizing specifically warm vs cold. The way this stuff shows up is as low Hb, period, in one of the below scenarios, where you just have to say, "Oh yeah, this is one of those conditions where you can get AIHA due to antibodies."	
Warm	- IgG antibodies against RBCs. - "Warm" because agglutination of RBCs occurs at warmer temperatures. The stem can say agglutination occurs when the nurse holds the tube of blood by hand. - Seen on USMLE for autoimmune diseases like SLE or RA; CLL; or can be drug- or infection-induced.
Cold	- IgM antibodies against RBCs. - "Cold" because agglutination of RBCs occurs at cooler temperatures. The stem can say agglutination occurs while the tubes are transported en route to the laboratory. - Seen on USMLE for *Mycoplasma* pneumonia and CMV mononucleosis.

Infectious mononucleosis
- Usually caused by EBV, but can also be CMV. - The virus invades B cells, which then stimulates the immune system to produce CD8+ T cells attacking the viral-infected B cells. In mono, these CD8+ T cells are called "atypical lymphocytes." The USMLE wants you to know these are **reactive CD8+ T cells.** They are "reactive" because they are responding to the viral-infected B cells. - Primary infection presents usually in teenager or young adult with fever >38 C, lymphadenopathy, tonsillar exudates, and lack of cough, making the presentation appear bacterial (in the HY Pulmonary PDF, I talk about CENTOR criteria for differentiating bacterial from viral URTIs). As a result, it is often misdiagnosed as *Strep* pharyngitis. - If amoxicillin or penicillin is given to treat EBV, this can cause a rash. This is not to be confused with a rash caused by allergy to beta-lactams. If pre-adolescent receives beta-lactam and gets a rash, that is likely beta-lactam allergy. If a patient adolescent or older gets a rash, we do a heterophile antibody (Monospot) test as next best step in diagnosis, as EBV mono is more likely. - Following the primary infection, mono can present as recurrent episodes of extreme fatigue that arise at interval-separations of months to years.

Langerhans cell histiocytosis (LCH)
- Overproduction of Langerhans cells, which are the dendritic cell of the skin. A dendritic cell is a strict antigen-presenting cell (APC), which phagocytoses antigen, migrates to lymph nodes and spleen, and presents it to CD4+ T cells. Macrophages and B-cells are also APCs, but they are not dendritic cells because they have additional functions – i.e., macrophages destroy antigens; B cells mature into plasma cells and produce antibodies. - The Langerhans cells that over-proliferate do so notably in bone, causing **lytic lesions** (e.g., of the skull). - Birbeck granules are tennis racquet- or rod-shaped inclusions within the Langerhans cells, visualized on electron microscopy. Students remember the granules are tennis racquet-shaped, but don't know the rod-shape. There is an LCH Q on NBME where they give the rod-shaped Birbeck granules, and students are confused because they say, "But wait, they're not tennis racquet-shaped though."

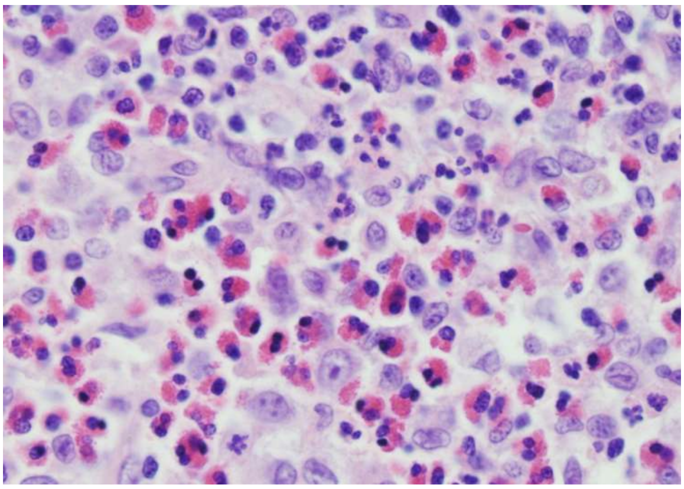


Tennis racquet-shaped Birbeck granules.

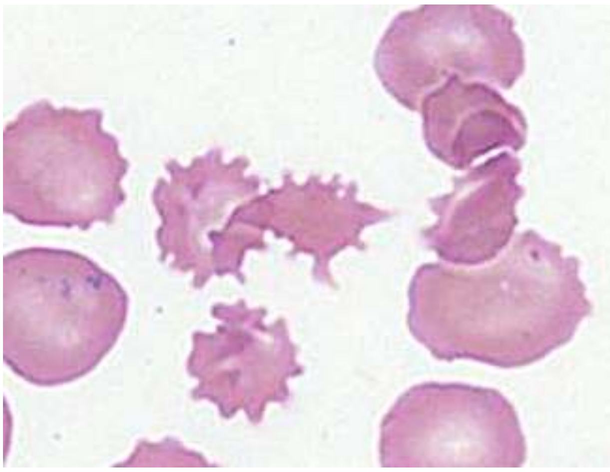


Rod-shaped Birbeck granules.

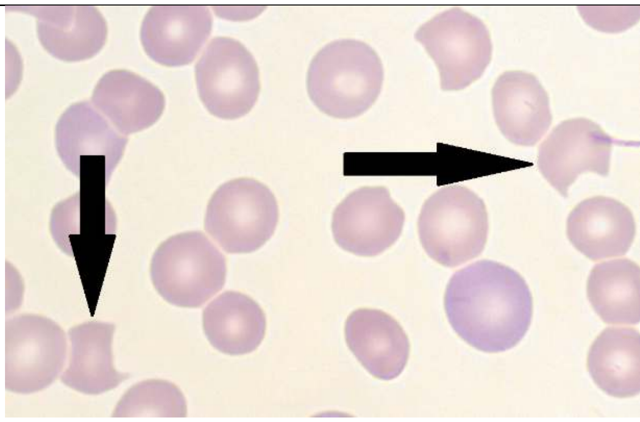
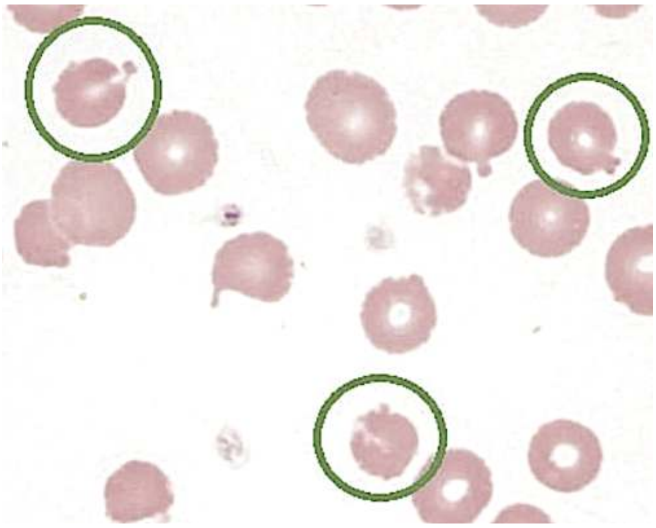
- There is difficult LCH NBME Q floating around where they don't show the Birbeck granules at all and instead just show the abnormal Langerhans cells straight-up. They will tell you a kid has a lytic lesion of his skull and then they show you image similar to the following:



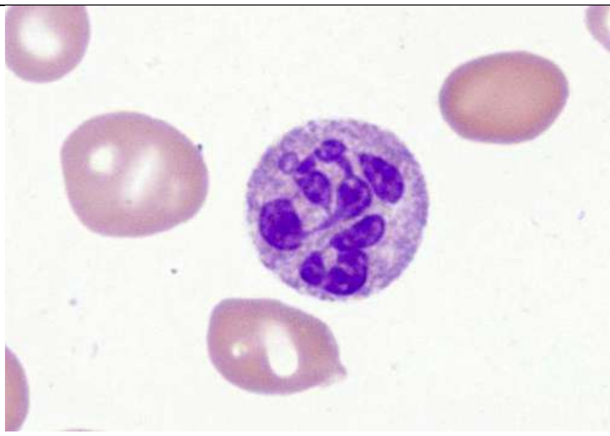
- You say, "No idea what I'm looking at." Agreed. But it's on the NBME. Apparently the Langerhans cells in LCH appear as large, irregularly shaped cells with copious eosinophilic cytoplasm and characteristic horseshoe- or kidney-shaped nuclei.

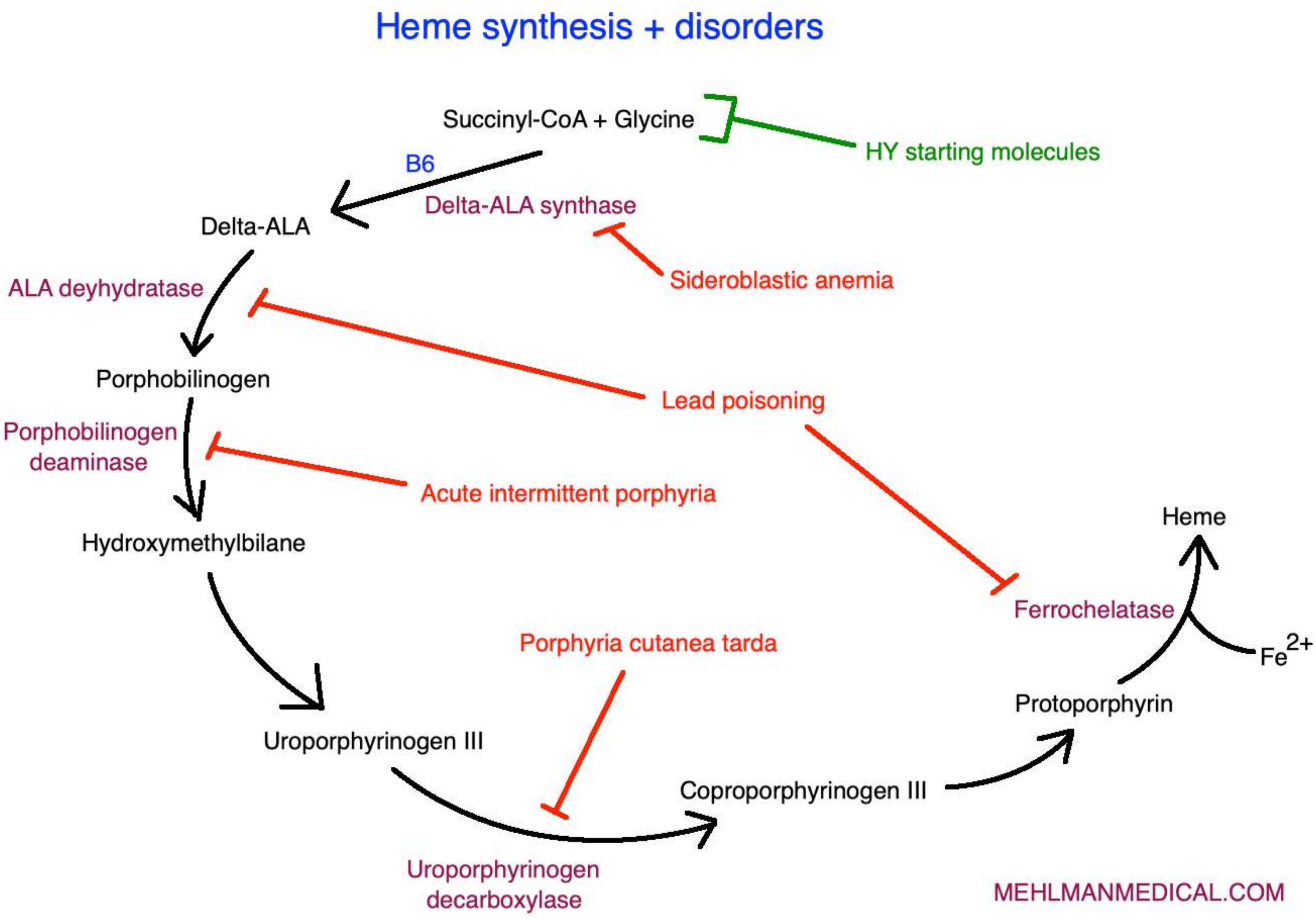
Acanthocytes (spur cells)
- Spiky RBCs that show up notably in **liver failure due to heat stroke.** - I've probably only seen around 4 questions with acanthocytes on NBME material ever. 3 out of 4 are liver failure; 1 out of 4 is abetalipoproteinemia. Other resources get the yieldness reversed, where they mention abetalipoproteinemia as though it's this HY association. Nonsense. USMLE cares about liver failure. - **Heat stroke** is end-organ damage due to hyperthermia (>104 F); heat exhaustion is mere fatigue with no end-organ damage due to hyperthermia. Heat stroke presents on Qs as abnormal liver and/or renal function tests. - Q will say old lady found in her house in the summer has body temp of 106 F and acanthocytes on a blood smear; what's the diagnosis? → answer = liver failure (due to heat stroke). - Another NBME Q gives hepatitis C patient with acanthocytes. The latter aren't critical to answer the Q, but I noted that the NBME mentioned them for hepatitis (i.e., not just heat stroke). 

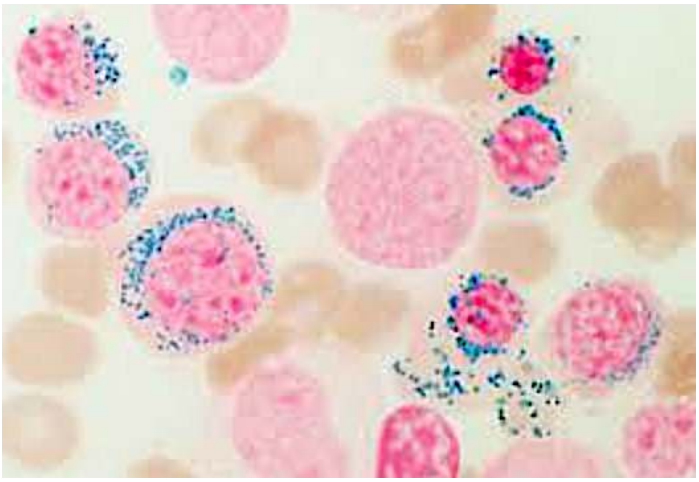
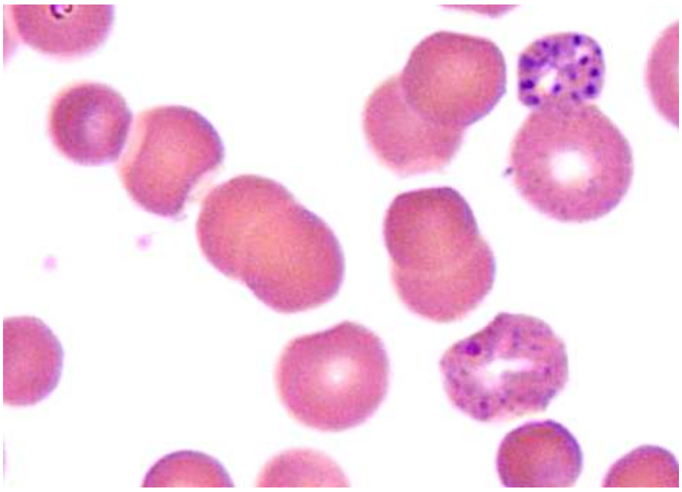
G6PD + PK deficiencies	
G6PD deficiency	- Glucose-6-phosphate dehydrogenase deficiency; **X-linked recessive.** - Most common cause of hemolysis due to an enzyme deficiency. - Presents as hemolysis and jaundice in boy who recently received a drug or had a viral infection. - HY agents that can precipitate oxidative damage of RBCs in G6PD are primaquine, dapsone, and sulfa drugs, and fava beans. - Vignette might be 12-year-old boy with scleral icterus following a viral infection. Labs might show increased unconjugated hyperbilirubinemia and LDH (due to hemolysis). RBCs are packed with LDH. High LDH can be the USMLE's way of saying there is hemolysis. - G6PD is an enzyme necessary to produce NADPH, which is a reducing agent that protects RBC membranes from oxidative damage. - Bite cells and Heinz bodies are buzzy and HY. Bite cells are RBCs with bite-like notches caused by oxidative damage. Heinz bodies are clumps of oxidized/denatured hemoglobin.

	 Bite cells  Heinz bodies
Pyruvate kinase	- Second most common cause of hemolysis due to an enzyme deficiency. - Leads to ↓ ATP production → ↓ activity of RBC Na/K-ATPase pumps → ↓ Na^+ pumped out of RBC → water stays with Na^+ → cellular swelling + lysis.

Macrocytosis causes	
- Normal RBC mean corpuscular volume (MCV) is 80-100 fL (femtoliters). - Macrocytosis refers to large RBCs (>100 fL).	
B9 (folate) / B12 deficiencies	- Deficiency of either vitamin leads to impaired DNA synthesis, where RBCs remain in the bone marrow for a longer time period, leading to RBCs that are larger and less mature. - High MCV as a result of impaired DNA synthesis is called megaloblastic anemia. - Impaired DNA synthesis also causes hypersegmented neutrophils, where longer time spent in the bone marrow results in an increased number of nuclear lobes and segments.

	
Alcohol	- An important cause of high MCV on USMLE in the setting of normal B9/B12 levels. - Alcohol can also cause other types of anemia (i.e., sideroblastic).
Orotic aciduria	- Obscure condition where DNA synthesis gets impaired, leading to increase in MCV. - Can be caused by various enzyme deficiencies. - Maybe shows up in one NBME Q in all of history. You could just peripherally be aware it's a rare as fuck cause of high MCV. - Leflunomide, an obscure drug used for rheumatoid arthritis that people used to memorize back during the numerical Step 1 era, inhibits dihydroorotate dehydrogenase, one of the enzymes that's deficient in orotic aciduria. Nonsense.



Heme synthesis disorders	
Sideroblastic anemia	- Impaired first step of heme synthesis (i.e., succinyl-CoA + glycine, via B6 and δ-ALA synthase $\rightarrow$ δ-ALA) due to deficiency or impairment of δ-ALA synthase. - Can be X-linked recessive, but also can be caused by alcohol. - Failure to incorporate iron into heme. This leads to "ringed sideroblasts," which are erythroblasts (RBC precursors) where iron accumulates inside the mitochondria, where the latter form a ring around the nucleus. - The iron-laden macrophages stain blue with Prussian blue stain, making this a very buzzy and pass-level image for USMLE. 
Lead poisoning	- USMLE wants you to know lead inhibits both δ-ALA dehydratase and ferrochelatase. - Results in buildup of δ-ALA. - The stereotypical vignette of a 2-year-old eating paint chips in an old house leading to mental deterioration is too easy and maybe only ~1/3 of lead questions. - The other 2/3 will be an adult who has **microcytic anemia** and mental decline. Buzzy hobbies can be hunting (bullets containing lead) or home-brewing of alcohol (casks made of lead).  - Can cause basophilic stippling of RBCs (rRNA precipitates). - Tx is calcium EDTA, dimercaprol, or succimer.
AIP	- Acute intermittent porphyria. - Autosomal dominant (sounds LY, but asked straight-up on an offline NBME somewhere). - Caused by deficiency of porphobilinogen deaminase (asked on NBME); causes buildup of porphobilinogen and δ-ALA. - Answer on USMLE for the heme synthesis disorder where patient has red/pink/port wine colored urine and abdominal pain.

	- Patients can sometimes have miscellaneous neurologic findings, but not mandatory. - Treatment is hematin and glucose.
PCT	- Porphyria cutanea tarda. - Caused by deficiency of uroporphyrinogen decarboxylase, leading to increased urinary uroporphyrins. - Answer on USMLE for the heme synthesis disorder where patient has photosensitivity / blistering. - Urine can appear "tea-colored" / dark. Don't confuse this with the red/pink/port wine colored urine in AIP. - What the USMLE will do is give you a vignette of PCT and then ask what is most likely the molecular starting point for this patient's disorder, and the answer is "succinyl-CoA" or "glycine," which reflects that this is a heme synthesis disorder.

Intra- vs extra-vascular hemolysis	
Intravascular	- Lysis of RBCs within the bloodstream. - Causes release of bilirubin directly into the blood, leading to **hemoglobinuria.** - **Serum haptoglobin is decreased.** Haptoglobin is a protein produced by the liver that binds to and removes free hemoglobin released from damaged or lysed RBCs. This mopping up of hemoglobin by haptoglobin prevents renal damage and helps to recycle iron. Low serum haptoglobin levels can be used to differentiate intra- from extra-vascular hemolysis). - A few HY examples of intravascular hemolysis are: - 1) G6PD deficiency (oxidative damage to RBCs). - 2) PNH, due to complement-mediated hemolysis; - 3) HUS, TTP, HELLP syndrome, DIC, mechanical hemolysis by prosthetic valves due to shearing of RBCs (all schistocytes).
Extravascular	- Phagocytosis and lysis of RBCs within the spleen (and also liver). - Hemoglobin from lysed RBCs is not released directly into the bloodstream; it is broken down within macrophages, forming biliverdin, which is converted to bilirubin. Bilirubin then goes to the liver for excretion in bile. - There is no hemoglobinuria. - Haptoglobin is not decreased. - HY examples are: 1) Hereditary spherocytosis (abnormal shape leads to clearance by spleen). 2) Thalassemias (imbalanced Hb chains leads to RBC structural imbalance, which is recognized by the spleen as defective RBCs). 3) Warm and cold AIHA, where Abs opsonize RBCs, leading to their removal in red pulp of the spleen. (It should be noted that Abs in AIHA can also lead to intra-vascular hemolysis if complement is recruited, but most destruction of RBCs in AIHA is extra-vascular.)

Hemolytic disease of newborn types	
Rhesus (Rh) type	- ~85% of people have Rh antigen on their RBCs – i.e., Rh(+). Women who are in the ~15% that are Rh(-) are at risk of developing IgG antibodies against Rh antigen if the fetus is Rh(+) and the circulations mix inadvertently. - When the mother develops antibodies against Rh (i.e., has become sensitized to it), we call this "Rh isoimmunization." NBME will throw this word around a bit, so you need to know it means "she's developed IgG antibodies against Rh." - Mechanism: Rh(-) mom in first pregnancy is exposed to fetal blood that is Rh(+); mom makes Abs against Rh; subsequent pregnancy results in IgG against Rh crossing placenta and targeting fetal Rh(+) RBCs, leading to hemolysis in the fetus.

	- As mentioned earlier, Rh status is checked at the first antenatal screening at 8-10 weeks. If the mother is Rh(-), then give RhoGAM at 28 weeks as well as at parturition. It must also be given if there are any interventions (e.g., amniocentesis), or if there's complications like spontaneous abortion or abruptio placentae. - If 2^{nd} pregnancy onward, if Rh (-) woman is found to have titers against Rh, do not give RhoGAM during the pregnancy, since it's too late. - Kleihauer-Betke (KB) test is done in setting of hemorrhage or suspected mixing of maternal and fetal circulations in Rh(-) women. The mother's blood is drawn, and the test works by exploiting the resistance of HbF to acid, allowing fetal RBCs to be distinguished from maternal cells on a blood smear. Depending on the fraction of fetal RBCs present, this determines the dose of RhoGAM necessary. - Another 2CK Q gives ↑ **fetal HR** in 3^{rd} trimester in woman who's in 2^{nd} pregnancy who has (+) "anti-D titers." They ask reason why fetal HR is ↑ answer = "Rh isoimmunization" → destruction of fetal RBCs → ↓ oxygen delivery within fetal circulation → ↑ HR to compensate.
ABO type **(2CK/3 only)**	- Most anti-A and -B Abs people have to opposing blood types are IgM, but people with O blood can have fractionally greater IgG. If mother with O blood has higher % of anti-A and -B Abs that are IgG, fetus can be symptomatic, since these will cross the placenta and attack the fetal RBCs. - USMLE will give you O+ mom usually in first pregnancy and A or B fetus that experiences pathologic jaundice. - ABO type HDN can theoretically occur in Rh- women in second pregnancies, but the USMLE wants to assess you specifically know the ABO type of HDN, so they'll give O+ mom in first pregnancy. In other words, they don't want you to get lucky choosing hemolytic disease of the newborn when you only know the Rh type.
Kell, Duffy, Kidd type **(2CK/3 only)**	- Minor antigens beyond Rh (i.e., Kell, Duffy, Kidd) can sometimes cause HDN. - The process for isoimmunization is same as Rh. - What will happen is, an A+/B+/AB+ mother will have a second pregnancy where fetal hemolysis is occurring, but based on her blood type, you know it's not ABO or Rh type HDN. - The Q will say her blood titer for, e.g., Kell is high. The next best step is "check Kell status of the father." - In other words, since the mother has titers against Kell, you know she's Kell negative. So if the father's RBCs are Kell(+), the fetus is likely Kell(+), and that would explain the HDN.

Transfusion reactions	
Febrile non-hemolytic transfusion reaction (FNHTR)	- Caused by activation of donor WBCs when they come into contact with antibodies in the recipient's plasma. - This interaction triggers the release of cytokines from donor WBCs. - These cytokines then cause symptoms of fever and chills, usually within a couple hours of the transfusion. This sets it apart from ABO mismatch, where symptoms usually occur within 10 minutes. - Coombs test is **negative,** since the process doesn't involve antibodies against donor RBCs. A (+) Coombs test means there are Abs against RBCs. - The vignette will be someone who gets a blood transfusion, and then within 2 hours gets fever + chills. They will then ask for the mechanism as the answer. I've seen them write it one of two ways: 1) Preformed antibodies against donor leukocyte antigens; and 2) Cytokines release from transfused blood.

	- Leukoreduction, a process that removes or reduces the number of WBCs from blood components before transfusion, can be used to mitigate risk of FNHTR by reducing probability of donor WBCs coming into contact with recipient antibodies.
Hemolytic (ABO mismatch)	- Caused by recipient antibodies against A and/or B antigens on donor RBCs. - This causes RBC agglutination and intravascular hemolysis, causing the release of hemoglobin into the bloodstream. - Binding of recipient antibodies to donor RBCs can also lead to complement activation and lysis of RBCs. - LDH can be increased in setting of hemolysis (RBCs are packed with LDH). - The vignette will be someone who gets fever, chills, and often flank pain within minutes of a blood transfusion. There can sometimes also be hypotension. - Coombs test is (+) because we have antibodies against RBCs. - Vignette can say, “10 minutes after the transfusion is started, the patient feels an ‘impending sense of doom’ and the transfusion is stopped” (sounds like MI, but NBME writes this). They then ask for next best step, which is antiglobulin test (another way of writing Coombs test). - FNHTR, in contrast, usually has symptoms within a couple hours, rather than within 10 minutes following the transfusion.
Delayed transfusion reaction	- Caused by recipient antibodies against minor antigens on donor RBCs, such as Rh, Kell, Duffy, Kidd. - Coombs test is (+), since we have antibodies against RBCs. - Vignette will be someone who had surgery + received RBCs intraoperatively + over the course of a week following the surgery, hemoglobin gradually decreases. The answer is then “Coombs (+); unconjugated bilirubin” as the combo that would be seen.
Transfusion-associated lung injury (TRALI)	- The answer on USMLE if a patient has an ARDS-like presentation with bilateral crackles and low O2 sats **<6 hours** following a transfusion. - Mechanism is abnormal priming of neutrophils in the lung that react to cytokines within transfused blood products (this is straight from a CMS Surg form 8 explanation). - Technically a type of non-cardiogenic pulmonary edema.
Transfusion-associated circulatory overload (TACO)	- Aka transfusion-induced hypervolemia. - Differs from TRALI in that this is a type of cardiogenic pulmonary edema (i.e., the left heart can’t handle the ↑ hydrostatic pressure from ↑ volume, so transudation into the alveoli occurs). - The answer on USMLE if a patient has an ARDS-like presentation **>6 hours** following a transfusion, OR gets ARDS-like presentation following transfusion in the setting of cardiovascular disease. In other words, presents one of two ways: 1) They don’t mention Hx of cardiovascular disease + the vignette will sound exactly like TRALI **> 6 hours** following a transfusion. - Surgery form 7 for 2CK gives an elderly dude who received only 3 packs of RBCs + he develops bilateral crackles and low O2 sats, but they say this occurs 12 hours after admission (not <6 hours as with TRALI), where the answer is “X-ray of the chest” as the next best step in diagnosis. The Q doesn’t rely on you discerning TACO vs TRALI to get it right, but the explanation says it’s TACO. 2) If a patient *with Hx of heart failure or MI* develops respiratory distress following transfusion of repeated blood products. - A 2CK NBME Q gives a 72-yr-old with Hx of MI ten years ago who gets shortness of breath and bilateral crackles 30 minutes after transfusion with crystalloid solution and 4 packs of RBCs (they don’t specify the volume of crystalloid in the Q).

	In summary, if ARDS-like picture after transfusion: If <6 hours → TRALI. If >6 hours → TACO. If heart disease → TACO regardless of time frame.

HY Heme/onc pharm points	
Aspirin	- Irreversible cyclooxygenase (COX) 1&2 inhibitor. - Non-steroidal anti-inflammatory drug (NSAID). - Aka salicylic acid (salicylate). - Blocks synthesis of prostaglandins and platelet thromboxane A2. - ↓ prostaglandin means ↓ inflammation, pain, and body temperature. - ↓ thromboxane A2 means ↓ platelet clumping. - Many HY use cases on USMLE: - Atrial fibrillation (with low CHADS2 score – i.e., Congestive heart failure, Hypertension, Age >75, Diabetes, Stroke/TIA/Hx of emboli, with the S being worth 2 points). If 0 or 1 points, give aspirin to AF patient; if 2+ points, give warfarin. USMLE doesn't obsess over valvular vs non-valvular AF in terms of giving warfarin vs NOACs (e.g., dabigatran, apixaban); they will either give aspirin or warfarin as the answer choices. - Peripheral vascular disease and carotid stenosis → USMLE wants triad of anti-platelet therapy (aspirin alone sufficient on USMLE), statin, and ACEi or ARB. - Ischemic heart disease (i.e., patients who have angina pectoris) → nitrates are used for acute angina, but daily aspirin is also given to these patients. - Ischemic strokes after 4.5 hours has transpired; if <4.5 hours, give tPA. - Aspirin can cause GI bleeds (↓ prostaglandin → ↓ gastric mucosal barrier mucous production and protection), nephropathy (pre-renal, interstitial nephropathy, nephrogenic diabetes insipidus), asthma (inhibition of COX causes shunting of arachidonic acid down the lipoxygenase pathway → leukotriene synthesis → bronchoconstriction). - Never give aspirin to children due to Reye syndrome risk (defect in beta-oxidation, with cerebral edema and hepatitis). - Misoprostol is given to patients following PPIs if they have NSAID-induced gastric ulcers (misoprostol is PGE1 analogue). - Can cause high-anion gap mixed metabolic acidosis-respiratory alkalosis (S in MUDPILES).
Other NSAIDs	- "Other NSAIDs" = reversible COX inhibitors, in contrast to aspirin (irreversible). - Indomethacin → used for acute gout and pericarditis; also closes PDA in neonates. - Ibuprofen → general NSAID used broadly for pain, fever, and autoimmune diseases (i.e., RA, IBD, SLE, etc.). - Naproxen → USMLE-favorite drug for causing GI bleeds and nephropathy; shows up on 2CK forms a lot. - Diclofenac, ketorolac (just know these names = NSAIDs). - 5-ASA compounds (mesalamine, sulfasalazine) used for IBD.
Acetaminophen	- Central-acting COX inhibitor. - Not considered an NSAID because it doesn't act peripherally. - Decreases pain and fever, but is not anti-inflammatory in the periphery. - First-line med for osteoarthritis (weight loss first Tx). NSAIDs not ideal first-line because they kill the kidneys + OA is non-inflammatory, so NSAIDs don't provide added benefit over acetaminophen. Patients with chronic pain who take NSAIDs frequently are at risk of nephropathy and GI bleeds. - Metabolite NAPQI causes massive hepatotoxicity. Activated charcoal can be given immediately after consumption, otherwise *N*-acetylcysteine HY as Tx. Latter regenerates reduced-glutathione, which neutralizes NAPQI.

Celecoxib	- Selective COX-2 inhibitor that does not cause GI bleeds. - Still can cause nephropathy. - NBME asks why celecoxib can increase the risk of MI → answer = decreases prostaglandin production without inhibiting platelets. - Can in theory be used for pain and inflammation.
Clopidogrel	- Anti-platelet agent; inhibits ADP P2Y12 receptor on platelets. MOA is HY. - Can be used to fulfill the anti-platelet component of the peripheral vascular disease and carotid stenosis triad (as mentioned before: 1) anti-platelet therapy; 2) statin; 3) ACEi or ARB). - In theory can be used for stenting procedures, but I haven't seen USMLE give a fuck. It's more that this use case is perpetuated in resources. - Can also be used in patients with ischemic heart disease who can't tolerate aspirin. - Prasugrel, ticlopidine, and ticagrelor drugs with same MOA, but not assessed.
Dipyridamole	- Phosphodiesterase-3 inhibitor; prevents breakdown of cAMP. - Causes arteriolar dilation + also an anti-platelet agent. USMLE really likes this – i.e., they will ask, "which of the following is both a vasodilator and anti-platelet agent?" → answer = dipyridamole or cilostazol (whichever is listed). - Dipyridamole/thallium is combo used frequently for pharmacologic stress testing. Arteriolar vasodilation causes HR to go up, where thallium uptake into myocardium can help visualize areas of ischemia.
Cilostazol	- Phosphodiesterase-3 inhibitor; prevents breakdown of cAMP. - Causes arteriolar dilation + also an anti-platelet agent, same as dipyridamole. - Used for intermittent claudication / peripheral vascular disease **after a walking/exercise program is already prescribed.** Do not choose cilostazol as a first treatment for peripheral vascular disease. Choose walking/exercise program first. Then, if they force you to choose a drug for additional management, choose cilostazol.
Abciximab	- GpIIb/IIIa inhibitor on platelets. - Prevents platelet aggregation (i.e,. platelet-platelet interaction). - Don't confuse with GpIb, which is for platelet adhesion. vWF normally bridges GpIb on platelets to vascular endothelium / underlying collagen, and is unrelated. - Since physiologically, fibrinogen is normally what bridges GpIIb/IIIa on adjacent platelets, abciximab is considered to be a fibrinogen analogue. - Drugs such as tirofiban and eptifibatide have same MOA but aren't assessed.
tPA (alteplase), streptokinase	- Tissue plasminogen-activator actives plasminogen, which breaks down fibrin clots. - I've only seen this used twice on NBME forms: 1) for ischemic strokes within 4.5 hours; and 2) for pulmonary embolism where there is obstructive shock (i.e., PE with low BP).
Heparin	- Activates anti-thrombin III, leading to inactivation of thrombin (factor II) and factors IXa, Xa, XIa, and XIIa. - Used as a bridging agent to warfarin when patient needs anticoagulation. This is because warfarin, by inhibiting protein C as part of its function, temporarily makes the patient hypercoagulable. - Causes both an ↑ in PT and aPTT, but aPTT is used to monitor heparin over PT because aPTT is more accurate/sensitive to the effects of heparin. This is because heparin inhibits intrinsic pathway factors (i.e., IXa, Xia, and XIIa) *in addition to* the common pathway factors (i.e., thrombin and Xa). In other words, 5 clotting factors within the intrinsic pathway are affected by heparin, but only 2 within the extrinsic pathway are (i.e., the common pathway ones). - Subcutaneous enoxaparin (LMWH) used first-line for DVT and superficial thrombophlebitis. Heparin is also used for acute limb ischemia and vertebral artery dissection (clot can form within false lumen; asked on 2CK NBME). - USMLE won't ask about unfractionated heparin vs LMWH, but apparently unfractionated heparin is used in patients with 1) severe renal impairment and 2) in those who might need rapid-reversal due to uncertainty about dosing. In other words, unfractionated heparin can be reversed with protamine sulfate much faster than LMWH.

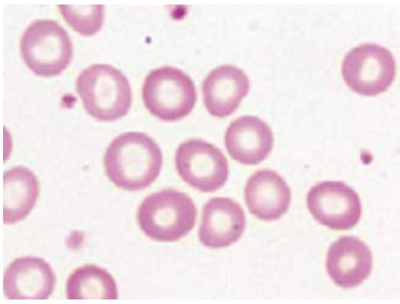
	- Heparin-induced thrombocytopenia (HIT) is exceedingly HY for USMLE. Mechanism is autoantibodies against heparin-platelet factor 4 (PF4) complex, where administration can lead to ↓ platelets. This is a type II hypersensitivity (platelets are cells). NBME can write the answer simply as "anti-platelet antibodies." Treatment is cessation of heparin + give direct-thrombin inhibitor (e.g., dabigatran). Warfarin is wrong answer. A 2CK NBME Q simply has "direct-thrombin inhibitor" as the answer, without listing specific drug name.
Warfarin, dicoumarol	- Vitamin K epoxide reductase inhibitors, which prevent vitamin K from being recycled back to its active form. - Vitamin K normally is a cofactor for an enzyme called γ-glutamyl carboxylase, which activates clotting factors II (prothrombin), VII, IX, and X, as well as anti-clotting proteins C and S. - PT is used for monitoring over PTT because many of the intrinsic pathway clotting factors (PTT) are unaffected by warfarin, whereas most of the extrinsic pathway clotting factors (PT) are affected by warfarin. - International normalized ratio (INR) is the ratio of the patient's PT to a control standard's PT, where 2.0-3.0 is the usual target range for warfarin. - Highest yield use cases on USMLE are for maintenance anticoagulation therapy in patients who have DVT, prosthetic valves, or atrial fibrillation with elevated CHADS2 score. - As discussed earlier, heparin must be used as bridging agent initially with warfarin because warfarin's inhibition of Protein C, which has a very short half-life, makes the drug initially pro-coagulable. - Warfarin levels are sensitive to drugs that inhibit or stimulate P-450. For example, P-450 inhibitors can ↑ INR and cause bleeding diathesis; P-450 inducers can ↓ INR and cause thromboses. - As discussed earlier, vitamin K deficiency due to broad-spectrum antibiotics causing depletion of bowel normal flora can ↑ INR (i.e., ↓ warfarin is required to inhibit clotting, but we haven't changed dose). - Teratogenic; must be avoided in pregnancy. Warfarin is a small, lipophilic molecule that readily crosses the placenta. In contrast, heparin is a large, acidic polymer that does not cross the placenta. Warfarin can cause bleeding diathesis and **bone abnormalities** in the fetus. - Can cause skin necrosis in patients with protein C deficiency. - Slowly reversed with vitamin K (requires re-synthesis of clotting factors by the liver); quickly reversed by fresh frozen plasma (supplies clotting factors).
Dabigatran	- Direct-thrombin inhibitor. - Used for HIT. - Sometimes in patients with non-valvular AF with high CHADS2 score.
Apixaban	- Factor Xa inhibitor. Just know MOA. - USMLE doesn't assess use cases, but can be given for non-valvular AF with high CHADS2 score.
Fondaparinux	- Factor Xa inhibitor. Just know MOA. - USMLE doesn't assess use cases, but can be given for DVT and PE in place of heparin in some patients.
Irinotecan, topotecan	- Topoisomerase I inhibitors.
Etoposide, teniposide	- Eukaryotic topoisomerase II inhibitors; anti-cancer agents. - Don't confuse with prokaryotic topoisomerase II (DNA gyrase) inhibitors, which refer to the fluoroquinolones (i.e., ciprofloxacin, levofloxacin).
Cisplatin	- Cross-linking agent. - Can cause nephro-, oto-, and neurotoxicity. - Chloride diuresis (i.e., normal saline) and amifostine can mitigate toxicity.
Doxorubicin (Adriamycin)	- Intercalating agent + produces free radicals. - Can cause dilated cardiomyopathy (HY). - Dexrazoxane mitigates toxicity by mopping up free radicals.

Bleomycin	- Induces DNA breaks. - Causes pulmonary fibrosis.
Carmustine, lomustine	- Alkylating agents.
Cyclophosphamide	- Guanine N7 alkylating agent. - Causes hemorrhagic cystitis (red urine). - Mesna mitigates toxicity by using a thiol (-SH) group to mop up acrolein, which is the toxic metabolite of cyclophosphamide that is toxic to uroepithelium.
5-fluorouracil	- Thymidylate synthase inhibitor. - Prevents conversion of dUMP to dTMP.
6-mercaptopurine	- Inhibits PRPP amidotransferase (enzyme in purine synthesis). - Requires xanthine oxidase for breakdown. - Allopurinol is a xanthine oxidase inhibitor that prevents tumor lysis syndrome in patients receiving chemotherapy for leukemia (lysis of leukemic cells releases nucleic acids into the blood, which are converted to uric acid). - Since 6-MP requires xanthine oxidase for breakdown, this means if patient is receiving 6-MP for chemotherapy, allopurinol cannot be given, otherwise 6-MP toxicity can result.
Azathioprine	- Pro-drug that is metabolized into 6-MP.
Mycophenolate mofetil	- Inhibits IMP dehydrogenase (an enzyme in purine synthesis). - Can be used for lupus nephritis.
Hydroxyurea	- Inhibits ribonucleotide reductase. - USMLE wants you to know it increases HbF in sickle cell to ↓ recurrence of painful crises. - Can be used to reduce bone marrow cell production in polycythemia vera to reduce hyperviscosity episodes. As discussed earlier, if USMLE gives you acute hyperviscosity syndrome, choose phlebotomy as best answer. Hydroxyurea is merely to decrease recurrence.
Microtubule drugs	- Vincristine, vinblastine, colchicine, griseofulvin, and -bendazoles (e.g., mebendazole) all inhibit microtubule polymerization. - Paclitaxel, docetaxel hyper-stabilize microtubules (i.e., prevent microtubule disassembly). - USMLE can ask for the protein these drugs interfere with → answer = tubulin. - Cells that are quickly dividing (i.e., GI mucosa, skin, hair follicles, bone marrow) are affected more than slowly dividing cells (e.g., myocardium). - If Q asks about the levels of which type of WBC are used to monitor for the toxic effects of chemotherapy, the answer is neutrophil (i.e., agranulocytosis). This is not limited to microtubule inhibitors, but I've seen USMLE ask this in relation to these drugs. - Vincristine is neurotoxic. - Colchicine is used for acute gout and pericarditis. - Griseofulvin is used orally for tinea capitis (patient only; don't give to close contacts; that distinction is asked on a 2CK FM form). - Mebendazole/albendazole are anti-helminthic agents, mostly for nematodes such as ascariasis, but can also be used for neurocysticercosis.
Imatinib	- Bcr/able tyrosine kinase inhibitor used to treat CML. - Can cause fluid retention / peripheral edema.
Erlotinib	- EGFR tyrosine kinase inhibitor used for non-small cell lung cancer.
Rituximab	- CD20 inhibitor on B cells. - Can ↑ risk of bacterial pneumonia (asked on NBME). This is because B cells mature into plasma cells, which make immunoglobulins, which are crucial for humoral immunity to fight bacterial infections.
Methotrexate	- Dihydrofolate reductase inhibitor (enzyme required in folate pathway). - First-line disease-modifying anti-rheumatic drug (DMARD) in rheumatoid arthritis. - First-line oral agent in plaque psoriasis that is non-responsive to topical agents, or for systemic psoriasis (arthritis).

	- Causes pulmonary fibrosis, mucositis (mouth ulcers due to neutropenia), and hepatotoxicity. - Toxicity mitigated with folinic acid (not folic acid), which is aka "leucovorin rescue."

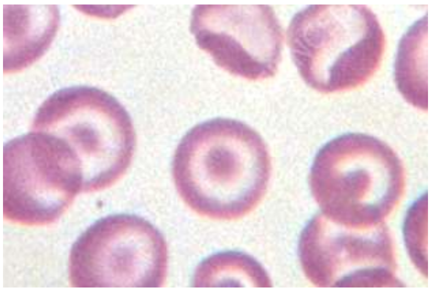
Secondary review (some students prefer bullet-point / vignette-style format). If you have already reviewed above tables, the following can be used to double-check your understanding of concepts. Otherwise you can move to another PDF.

- 32F + low MCV + low Hb + smear is shown below; Dx?



  - Answer = iron deficiency anemia (IDA); smear shows pale RBCs (i.e., with central pallor).

- 32F + low MCV + low Hb + smear is show below; Dx?



  - Answer = thalassemia; smear shows target cells; these may be seen in other DDx, however on USMLE thalassemia is by far the highest yield association.

- 32F + low MCV + low Hb; next best step in Mx? → answer = "check serum iron and ferritin."
- 32F + low MCV + low Hb + low Fe + low ferritin; Dx? → answer = IDA.
- 32F + low MCV + low Hb + normal Fe + normal ferritin; Dx? → answer = thalassemia. Presents as a microcytic anemia *despite* a normal iron and ferritin.
- 32F + normal MCV + low Hb + low Fe + normal ferritin; Dx? → answer = anemia of chronic disease (AoCD). In contrast, IDA you'll have low iron + low ferritin; thalassemia you'll have normal iron + normal ferritin. Also, for AoCD, there will be obvious cause like renal failure, autoimmune disease, cancer, or chronic infection like hepatitis B/C.

- 32F + rheumatoid arthritis + low MCV + low Hb + low Fe + normal ferritin; Dx? → answer = AoCD; student says, "Wait, but I thought MCV was supposed to be normal in AoCD." → plenty of 2CK-level Qs give **low** MCV for AoCD. That's you're value point.
- 32F + low MCV + low Hb + high red cell distribution width (RDW); Dx? → answer = IDA almost always. One question in history gives high RDW for thalassemia (on FM form 4). But usually this is seen in IDA.
- 32F + low MCV + low Hb + low/normal RDW; Dx? → answer = thalassemia.
- 32F + low MCV + low Hb + high transferrin binding capacity; Dx? → answer = IDA.
- 32F + low MCV + low Hb + low/normal transferrin binding capacity; Dx? → AoCD.
- 32F + low MCV + low Hb + no improvement with iron supplementation; Dx + next best step in Mx? → answer = thalassemia; next best step = hemoglobin electrophoresis.
- Mechanism for thalassemia? → decreased production of one type of hemoglobin chain (i.e., if ↓ alpha chain production, then Dx is alpha thalassemia; same for beta, respectively).
- 32F + pregnant + completely asymptomatic till this point + low MCV + low Hb + starts taking prenatal vitamin supplement + three weeks later still has low MCV and low Hb; next best step? → answer = hemoglobin electrophoresis. Dx = thalassemia.
- 32F + pregnant + Hx of chronic fatigue + low MCV + low Hb + normal iron and ferritin; Dx? → answer = alpha thalassemia trait (two mutations) or beta-thalassemia minor; patient will be adult who presents similar to mild/moderate IDA but have normal serum iron and ferritin.
- 8F + severe anemia + hepatosplenomegaly + normal iron and ferritin + Hb electrophoresis shows tetrameric beta-hemoglobin (β4); Dx? → answer = hemoglobin H disease (three alpha mutations).
- 32F + pregnant + fetus dies *in utero* + fetal blood sampling shows tetrameric gamma-hemoglobin (χ4); Dx? → hemoglobin Barts (four alpha mutations); fatal *in utero*.
- 6M + low MCV + low Hb + hepatosplenomegaly + HbA2 6% + following x-ray of head; Dx?

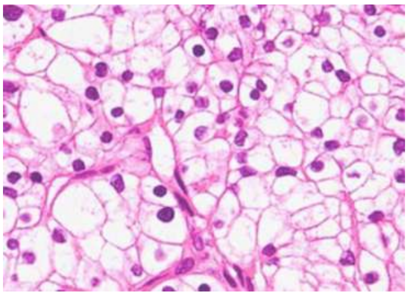
- Answer = beta-thalassemia major; almost always a child; "chipmunk facies/skull" and hepatosplenomegaly occur due to ↑↑ extramedullary hematopoiesis.

- 6M + low MCV + low Hb + normal iron and ferritin + hepatosplenomegaly + HbA2 6%; Tx? → answer = serial blood transfusions + iron chelation therapy (e.g., deferoxamine) for beta-thalassemia major; serial blood transfusions done to Tx thalassemia result in iron overload; do not confuse this type of iron overload with that of hereditary hemochromatosis, which is instead managed with serial phlebotomy, not chelation therapy. The iron overload due to serial transfusions is called secondary hemochromatosis (transfusional siderosis).
- 10F + receiving transfusions for beta-thalassemia major; Q asks: to avoid iron overload, measurement of which of the following is most sensitive in assessing patient's iron stores; answer = ferritin; wrong answer are iron saturation and transferrin.
- 22F + low MCV + low Hb + fatigue + HbA2 6%; Dx? → beta-thalassemia minor; usually adult.
- "What's the deal with the HbA2. What does that mean?" → highest yield point for beta-thalassemia major and minor is that HbA2 ($\alpha2\delta2$; alpha2-delta2) is increased on hemoglobin electrophoresis. Normal adult Hb is HbA1 ($\alpha2\beta2$); fetal Hb is HbF ($\alpha2\chi2$; alpha2-gamma2).
- Normal MCV? → 80-100 fL.
- Normal Hb? → according to literature, in men + elderly it's 13.0-17.5 g/dL; menstruating females 12.0-17.5 g/dL. Below these thresholds, patient has anemia. Above, patient has polycythemia.
- Normal Hct? → Men and elderly: 42-52%; women: 36-46%.
- Normal WBC? → 4-11,000 / uL.

MEHLMANMEDICAL.COM	**Iron deficiency anemia**	**Thalassemia**
Hb	↓	↓
MCV	↓	↓
Serum iron	↓	Normal

Ferritin	↓	Normal
RDW	↑ (HY)	↓/Normal (HY)
Blood smear	Pale RBCs	Target cells
Hb electrophoresis	Normal	α: normal; **β: ↑ HbA2 (HY)**
What happens if we give iron?	Improvement	**No improvement (HY)**
Tx	Iron	Transfusions if severe

- 72M + fatigue + smear shows pale RBCs + Hb 9.4 g/dL; most likely cause? → answer = GI blood loss (IDA) → must think diverticular bleed, colorectal cancer, and angiodysplasia causing IDA in elderly patient with fatigue; 2CK-level Qs will jump straight to colonoscopy as the answer.
- 65M + pain in fingertips for 3 weeks + facial plethora + splenomegaly + Hb 20.2 g/dL + WBCs 14,500 with normal differential + normal platelets + O2 sats 94% on room air; Dx + Tx? → answer = polycythemia vera (PCV); Tx = phlebotomy.
- Mechanism of PCV? → JAK2 mutation causing "proliferation of bone marrow stem cells." Erythropoietin (EPO) is **decreased** because it is suppressed. Although oxygen sats should be as close to 100% as possible, patients generally hold up fine with sats >94%.
- 48F + pruritis after a shower + high WBCs + Hb 19.5 g/dL; Dx? → answer = PCV; pruritis after a shower is a classic finding → reflect basophilia; WBC count can be normal or elevated in PCV.
- 50F smoker + Hx of COPD + Hb 18.5 g/dL; Dx + mechanism? → answer = secondary polycythemia → **increased** EPO due to low oxygen tension (e.g., in COPD, CF, etc.) → mechanism is "proliferation of erythroid precursors" → this is because the high EPO results in an elevation of *only* RBCs; in PCV, O2 sats would be 94% or greater and WBCs and/or platelets may be elevated.
- 55M smoker + red urine + polycythemia + hypercalcemia + unknown biopsy specimen is shown; Dx?



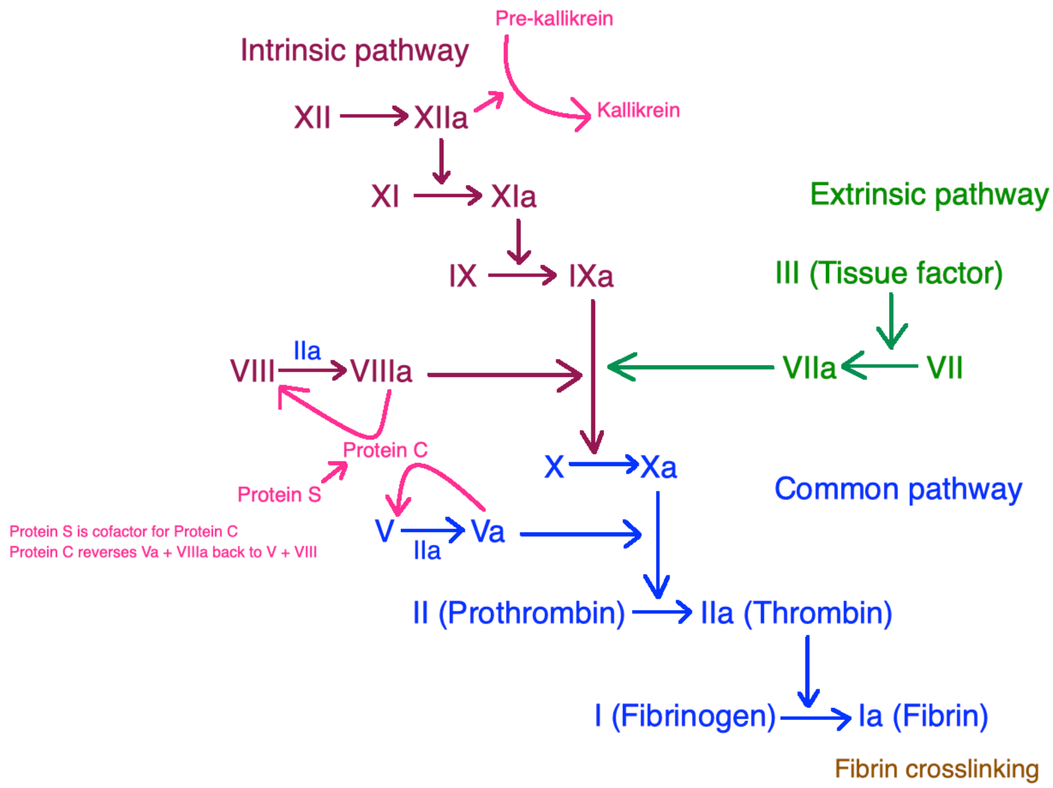
  - Answer = renal cell carcinoma (RCC) → can cause paraneoplastic secondary polycythemia due to EPO secretion as well as hypercalcemia due to PTHrp secretion (latter also squamous cell of lung); biopsy shows clear cell carcinoma (most common variant of RCC; HY biopsy finding for Step 1).

- Normal bleeding time (BT)? → answer = 2-7 minutes.
- Normal platelet count (PC)? → answer = 150-450,000/uL.
- Two main ways ITP presents on USMLE? → 1) school-age kid with viral infection followed by epistaxis and/or bruising/petechiae; 2) woman 30s-40s with random bruising and elevated BT / low platelets.
- 34F + bruises + BT 9 minutes; Dx? → answer = idiopathic/immune thrombocytopenic purpura (ITP).
- 34F + bruises + BT 6 minutes; Dx? → answer = domestic abuse (on FM shelf).
- 34F + viral infection followed by petechiae + epistaxis; which would confirm the Dx (answers are either ↑ bleeding time or ↓ platelet count) → answer = ↓ platelet count; both are seen, but an isolated ↓ platelet count in the setting of the clinical picture is how to confirm Dx of ITP.
- 12M + runny nose for 4 days + epistaxis; Dx? → answer = ITP → often follows viral infection.
- 12M + no mention of any type of infection + epistaxis + BT 8 minutes; Dx? → answer = ITP → student says, "Wait, but you just said it follows a viral infection." → **USMLE need not mention viral infections, and they often don't.** Especially on 2CK-level NBME Qs, for conditions associated with viral infections like ITP, deQuervain subacute granulomatous thyroiditis, minimal change disease, IgA nephropathy, etc., the Q won't even mention viral infection; you just need to know the presentation. Just think: most people with COVID-19 are/were asymptomatic right?
- 12M + cough + coryza + epistaxis; mechanism for this patient's condition? → answer = **antibodies against glycoproteins IIb/IIIa on platelets** (type II hypersensitivity); GpIIb/IIIa mediate platelet aggregation (not adhesion, which is GpIb, not IIb/IIIa); there is often a genetic susceptibility to ITP. This is asked on 2CK material as well.
- 12M + epistaxis + platelet count 50,000; next best step in management? → answer = steroids.
- 12M + epistaxis + platelet count 50,000 + steroids not effective; next best step in management? → answer = IVIG.
- 12M + epistaxis + platelet count 50,000; what's the most effective way to decrease recurrence? → answer = splenectomy; students says, "Wait, I thought you just said steroids were what we do first." → Yeah, you're right, but the Q asks what's **most effective** in decreasing recurrence, so splenectomy is correct. USMLE will sometimes ask next best step vs what's most effective.

- Two points to note about ITP Tx:
  - 1) some literature has suggested no Tx is necessary in select pediatric cases where the presentation is limited to mild cutaneous findings. However on the USMLE, steroids is always the answer for immediate management.
  - 2) Literature suggests sometimes IVIG may be given before steroids (i.e., extremely low platelet counts). But on the USMLE, steroids are always first, not IVIG.

- 12M + viral infection + low neutrophils; Dx? → answer = viral-induced neutropenia (not ITP).
- 12M + viral infection + low neutrophils + fever; next best step in Dx + Mx? → answer = neutropenic fever (febrile neutropenia) → medical emergency; must give immediate IV broad-spectrum antibiotics; patient has possible infection but no way to defend against it.
- 12M + low neutrophils + fever + antibiotics are administered; what could help restore this patient's neutrophil count? → answer = granulocyte-macrophage colony-stimulating factor (GM-CSF); molgramostim is commonly used as the GM-CSF agent.
- 8M + viral infection + low Hb + low WBCs + low platelets; Dx? → answer = aplastic anemia (viral induced; likely Parvo-B19); aplastic anemia = Dx when all cell lines are down due to decreased bone marrow production).
- 32F + malar rash + arthritis + Hb 9 + WBCs 3000 + platelets 90,000; mechanism for this hematologic presentation? → answer = "increased peripheral destruction" (i.e., autoantibodies); "decreased bone marrow production" is the **wrong** answer in SLE; lupus is associated with the development of anti-hematologic cell line antibodies, usually against platelets (i.e., it's common to see isolated thrombocytopenia in SLE), but antibodies can form against WBCs and RBCs as well. What makes this difficult is that a ↓ in all cell lines (i.e., RBCs, WBCs, platelets) looks like aplastic anemia (i.e., such as with Parvo), but with SLE, the finding is due to antibodies, **not** defective bone marrow production. In contrast, if this were viral-induced, then yes, that is aplastic anemia, and the answer is "decreased bone marrow production."
- 44F + hospitalized and treated for overactive thyroid + Hct 45% + WBCs 1500 (neutrophils 5%; lymphocytes 95%) + PC 250,000; Dx? → answer = drug-induced neutropenia → propylthiouracil and methimazole both can cause agranulocytosis (neutropenia); other HY drugs for agranulocytosis are ganciclovir (for CMV), clozapine (anti-psychotic), methotrexate (DMARD), ticlopidine (anti-platelet).

- 3M + absent thumb on the left hand + ↓ Hb + ↓ WBCs + ↓ platelets; Dx? → answer = Fanconi anemia; autosomal recessive aplastic anemia; presents with absent or hypoplastic thumbs or radii.
- 3M + triphalangeal thumb + ↓ RBCs; Dx? → answer = Diamond-Blackfan anemia; pure RBC aplasia that presents with triphalangeal thumb.
- Cancer associated with pure RBC aplasia? → answer = thymoma.
- 8M + viral infection + low Hb + low WBCs + low platelets + fever; next best step? → answer = immediate IV antibiotics → even though viral-induced aplastic anemia, the patient *still* has a neutropenic fever, so this is a medical emergency.
- 8M + viral infection + low Hb + low WBCs + low platelets; what's the next best step in Dx? → answer = bone marrow aspiration (to confirm decreased bone marrow production consistent with aplastic anemia); sounds overkill, but it's the answer.
- 26F + daycare worker + coryza + lacy rash on legs and trunk + all immunizations up to date + afebrile + RBCs, WBCs, and platelets all normal; next best step in Dx? → answer = "check Parvovirus IgM titers" → in the absence of aplastic anemia, this is the answer if daycare worker presents with rash.
- 44F + undergoing chemotherapy + low RBCs + low WBCs + low platelets; next best step in Dx? → answer = bone marrow aspiration (chemo-induced aplastic anemia).
- 44F + undergoing chemotherapy + all cell lines down + temperature 101.8F; next best step? → answer = immediate IV broad-spectrum antibiotics; this is chemo-induced aplastic anemia, but there's a neutropenic fever.
- "Do we have to know the clotting cascade?" → To understand many heme disorders, yes, you should know it:



- "Can you explain PT and PTT in relation to extrinsic and intrinsic pathways?"
  - PT reflects the functioning of the extrinsic pathway; PTT reflects intrinsic. In other words, if PT alone is high, then the extrinsic pathway is fucked up. If PTT alone is high, then the intrinsic pathway has a problem. If both PT and PTT are elevated, then the common pathway has an issue.
- Normal prothrombin time (PT)? → answer = 10-15 seconds.
- Normal activated partial thromboplastin time (aPTT; PTT)? → answer = 25-40 seconds.
  - aPTT is a slightly more sensitive version of PTT, but on USMLE they are used interchangeably.
- 55F + no Hx of bleeding problems + coagulation testing prior to CABG shows ↑ PTT + normal PT + normal BT; what's most likely to be abnormal in this patient? → answer = USMLE answer is kallikrein formation. Patients with ↓ factor XII (Hageman factor) are asymptomatic; factor XII converts pre-kallikrein into kallikrein.
- Neonate + born at home + bleeding from umbilical site; what are the arrows for BT, PT, and aPTT? → answer = normal BT; ↑ PT; ↑ aPTT; Dx = vitamin K deficiency. Born at home = not given vitK injection.

- 40M + on warfarin for prosthetic valve placed years ago + receiving broad-spectrum antibiotics for infection; Q asks why required warfarin dosage over next several weeks would ↓; answer = "vitamin K deficiency caused by depletion of normal gut flora."
- What does vitamin K do? → cofactor for enzyme called gamma-glutamyl carboxylase → gamma-carboxylates + activates clotting factors II, VII, IX, and X, as well as anti-clotting proteins C and S. Since factors II (prothrombin) and X are in the common pathway, PT and PTT are both elevated in vitamin K deficiency. Factor VII is in the extrinsic pathway; factor IX in the intrinsic pathway. Protein C functions to inactivate factors Va and VIIIa (activated clotting factors V and VIII) back to their inactive form. Protein S is merely a cofactor for protein C.
- How does warfarin relate to vitamin K? → Warfarin inhibits vitamin K epoxide reductase, which is the enzyme that recycles vitamin K to its active form. That means less vitamin K can act as a cofactor for gamma-glutamyl carboxylase. Notice the enzymes are different. Therefore warfarin leads to decreased activation of clotting factors II, VII, IX, and X, as well as anti-clotting proteins C and S.
- "If proteins C and S are anti-clotting, and warfarin inhibits their activation, why would warfarin be an anticoagulant then?" → protein C has a super-short half-life, so the effect of warfarin in the first few days actually results in a hypercoagulable state, where protein C is ↓ but the actual clotting factors are still present in greater amounts; this is why heparin is necessary as a bridging agent – i.e., patients commenced on warfarin require simultaneous commencement of heparin for a few days.
- What does heparin do? → activates antithrombin III → leads to inactivation of prothrombin (factor II) and factor X.
- How do you reverse warfarin? → vitamin K (slow; takes several days); if patient is actively bleeding or requires surgery (i.e., fast reversal), answer = fresh frozen plasma.
- How do you reverse heparin? → protamine sulfate.
- Anything important about the structures of heparin vs warfarin? → heparin is a large, acidic, anionic molecule and therefore does not cross the placenta; protamine, which is a positively charged cation, can bind to and chelate it. Warfarin is a small lipophilic molecule that can cross the placenta, and is therefore teratogenic.

- 34F + DVT + PC 220,000 + PT 13 seconds + PTT 36 seconds + heparin commenced + now PC is 130,000; mechanism? → answer = drug-related antibodies → heparin-induced thrombocytopenia (HIT) → type II hypersensitivity → antibodies form against the heparin-platelet factor 4 complex.
- Tx for HIT → stop heparin and give direct-thrombin inhibitor (e.g., dabigatran, lepirudin); warfarin is the number-one wrong answer.
- How does a platelet problem present? → epistaxis or generally mild cutaneous findings (i.e., petechiae, bruising).
- How does a clotting factor problem present? → menorrhagia; excessive bleeding after tooth extraction; hemarthrosis; excessive bleeding after circumcision (neonatal boys).
- 17F + Hx of epistaxis + sometimes clots in her menstrual pads; Dx till proven otherwise? → answer = von Willebrand disease (vWD) → always presents with combination of platelet problem + clotting factor problem. Clots with menses signify heavy bleeding (i.e., menorrhagia).
- Inheritance pattern of vWD? → **autosomal dominant (high-yield).**
- Mechanism of von Willebrand factor? → bridges GpIb on platelets to underlying collagen + vascular endothelium → mediates platelet **adhesion** (don't confuse this with platelet aggregation, which is when platelets stick together via GpIIb/IIIa); this leads to increased bleeding time. vWF also has secondary/ancillary function of stabilizing factor VIII in plasma → in vWD, **PTT is elevated in ~1/2 of question stems** (students tend to erroneously memorize PTT as always ↑, but this will get you questions wrong).
- 17F + Hx of epistaxis + Hx of excessive bleeding with wisdom teeth removal + BT 9 minutes + PT 12 seconds + PTT 44 seconds; Dx? → answer = vWD → BT always ↑; **PTT ↑ in only ~half of questions.**
- 17F + Hx of epistaxis + cut her finger a month ago that took a long time to stop bleeding + PT normal + PTT normal + platelet aggregation studies normal; Dx? → vWD → the ↑ BT is reflected by the cut on the finger taking a while to stop bleeding; PTT need not be elevated; normal platelet aggregation studies simply mean that GpIIb/IIIa are functioning properly, but vWD relies on GpIb, which is adhesion, not aggregation.
    - When students get above vignette wrong, it's because they don't realize PTT is normal in about half of vWD vignettes and/or they forget vWF mediates platelet adhesion, not aggregation.

- 19M + petechiae + normal platelet count + BT 9 minutes + PTT 42 seconds; Tx for this patient's condition? → answer = desmopressin (DDAVP) increases vWF synthesis and release.
- 16F + took aspirin in suicide attempt + ↓ Hb + blood in stool; Q asks what finding would be expected to be abnormal in this patient? → answer = bleeding time (↑↑ because aspirin inhibits COX1/2 → ↓ thromboxane A2 production in platelets → ↓ platelet function); the wrong answer is ↓ platelet count (platelet count doesn't change with aspirin/NSAID use); fibrin degradation products is also a wrong answer (this is ↑ in DIC and pulmonary embolism).
- 53M + coronary artery disease + asks physician about celecoxib + physician is reluctant to prescribe because of increased risk of MI with celecoxib; Q asks why there is ↑ risk; answer = "inhibition of prostacyclin (PGI2) formation without inhibition of thromboxane A2 in platelet"; celecoxib is a COX2 selective inhibitor.
- 8M + hemarthrosis + PTT 90 seconds; Dx? → answer = hemophilia A or B → isolated ↑ in PTT.
- Hemarthrosis in school-age boy; Dx till proven otherwise? → answer = hemophilia A or B.
- Hemarthrosis in school-age boy + no way to differentiate between A and B based on the vignette; Dx? → answer = hemophilia A (way more common than B); for some reason, this epidemiologic point is assessed on the Step in this fashion.
- 8-day-old neonatal male + excessive bleeding with circumcision; Dx? → answer = hemophilia A or B.
- Inheritance pattern of hemophilia A and B? → **X-linked recessive (high-yield).**
- Mechanism for hemophilia A and B? → answer = deficient production of factors VIII (A) and IX (B). Rarely, they can be caused by antibodies against the factors.
- Tx for hemophilia A? → answer = IV desmopressin (DDAVP) and/or factor VIII replacement.
- Tx for hemophilia B? → answer = factor IX replacement only.
- 15M + Hx of receiving factor VIII replacement therapy for hemophilia A + becoming increasingly less effective with time + PTT is 160 seconds; Dx? → answer = antibodies against factor VIII (almost always due to repeated replacement of factor VIII).
- 13F + PTT 90 seconds + analysis shows deficiency of factor IX; Q asks the mechanism via which this is possible → answer = lyonization (skewed X-inactivation); the USMLE Q will never give a female with

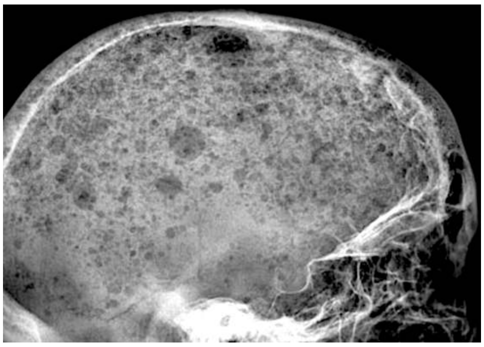
an X-linked recessive disorder unless the explicit point of the Q is X-inactivation/lyonization. That is, never assume, “hmm well maybe there’s lyonization here…” If the USMLE wants that, they’ll ask it.

- 23M + BT 12 minutes + thrombocytopenia + smear shows giant platelets + failure of platelet agglutination with ristocetin cofactor; Dx + mechanism? → answer = Bernard-Soulier disease; mechanism = deficiency of platelet GpIb (mediates adhesion); platelets can be giant for some magical reason.
- “Wait what’s that ristocetin assay thing. I’ve seen that before.” → all you need to know is that it will cause platelet agglutination, **but in Bernard-Soulier disease and vWD, the assay is negative** – i.e., the platelets don’t agglutinate. The test measures the binding of vWD to platelet GpIb, so clearly deficiency of either will yield a negative test.
- 23M + BT 12 minutes + normal platelet count + ristocetin cofactor assay yields agglutination; Dx + mechanism? → answer = Glanzmann thrombasthenia; mechanism = deficiency of platelet glycoproteins IIb/IIIa (mediate aggregation, not adhesion). Student says, “Wait, but if it’s an isolated increase in BT, why can’t this just be ITP then?” → Because platelet count is normal; in ITP, there’s always thrombocytopenia. I’ve seen this Dx on the NBMEs show up as just “thrombasthenia,” where you’re like “Wtf is thrombasthenia?” But that just refers to Glanzmann.
- 32F + Hx of Crohn + low Hb + MCV 90; Dx? → answer = AoCD → seen in autoimmune disease (i.e., RA, IBD, SLE), organ failure (i.e., renal, liver), chronic infection (e.g., hepatitis B/C).
- Mechanism of AoCD? → inflammatory state leads to ↑ IL-6 production by liver → ↑ hepcidin production by liver → ↓ ferroportin acivity → ↓ iron release by gut enterocytes and general cellular stores → ↓ iron transport in blood → ferritin levels are normal but serum iron is ↓. Transferrin is also ↓, resulting in ↓ total iron binding capacity (TIBC). This means even though iron is low in the blood, since transferrin is also low, there’s still ↓ binding capacity for iron overall. In contrast, in IDA, TIBC is high (i.e., transferrin goes ↑ to compensate for low serum iron; in AoCD, transferrin secretion is suppressed despite ↓ serum iron).
    - Student says, “But wait, isn’t this similar to thalassemia, where serum iron is decreased but ferritin is normal?” → Good question, but in thalassemia, 1) they won’t give you a vignette

with autoimmune disease, organ failure, or chronic illness, and 2) they'll often throw in target cells, ↓ RDW, or Hb electrophoresis findings.

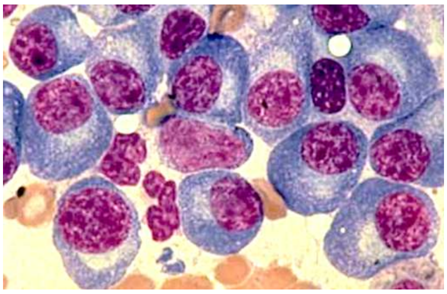
- "What about MCV and AoCD?" → classically normal MCV in anemia of chronic disease (normocytic anemia), but some 2CK NBME Qs have ↓ MCV → student says, "Wait how is this AoCD? Isn't MCV supposed to be normal in AoCD?" → my response: "Yes, you're right, but various 2CK have it ↓." → So your take-home regarding AoCD should be: classically normal, but can absolutely be ↓ on USMLE.
- 6F + Hx of multiple episodes of sore joints + fever of 102F + salmon-pink rash over body + high ESR + Hb of 9 g/dL + MCV 72; Dx? → anemia of chronic disease secondary to juvenile rheumatoid arthritis (JRA; aka juvenile idiopathic arthritis; JIA) → if Q gives you low MCV in AoCD, the vignette will be overwhelmingly obvious for an autoimmune disease + you can eliminate the other answers.
- 44F + chronic alcoholism + abdominal fluid wave + low Hb + MCV 84; Dx? → AoCD.
- 68M + hasn't been to doctor in years + Hb 8.6 + Hct 25% + MCV 90 + normal RDW + normal iron + normal ferritin + normal transferrin saturation + creatinine 2.9; Dx + Tx? → AoCD caused by chronic renal insufficiency. Tx = EPO.
- Tx of AoCD? → answer = supportive care / treat the underlying condition; **EPO is the answer only if renal failure is the etiology; if renal failure is not the underlying Dx, EPO is the wrong answer.**
- 12F + chronic renal failure + epistaxis for past two weeks + platelets 200,000/uL + Hb 9.5 g/dL; Q asks the mechanism for the epistaxis; answer = "acquired platelet dysfunction"; Dx = uremic platelet dysfunction → high blood urea nitrogen (BUN) in renal failure causes a **qualitative** dysfunction of platelets, where they merely don't do their job; there is **no quantitative** issue (i.e,. platelet count is normal).
    - Students often choose "erythropoietin deficiency," which is the wrong answer. Anemia of chronic disease can be seen in renal failure secondary to decreased EPO, yes, but AoCD in and of itself doesn't cause epistaxis; epistaxis is seen with platelet problems; Hb is merely down because there's IDA from the epistaxis.
- 82F + back pain + M-protein spike showing IgG kappa + epistaxis; why the epistaxis? → uremic platelet dysfunction secondary to renal failure from multiple myeloma (renal amyloidosis).
- "Can you tell me the highest yield points about multiple myeloma?"

- Cancer of **plasma cells** → monoclonal expansion, where the bone marrow has ↑↑ plasma cells (>10%) originating from a single plasma cell.
- Plasma cells secrete non-functional immunoglobulin light-chains; these are mostly IgG; **serum protein electrophoresis** is the next best step in the Dx of multiple myeloma, which will show the ↑↑ IgG kappa and lamda; this ↑↑ in IgG is called an M-protein spike; this has nothing to do with IgM; don't confuse that.
- Since the IgG light chains are proteins and are present in ↑↑ amounts in the blood, they are prone to deposit in tissues, leading to **amyloidosis** (abnormal protein deposition in tissues)**;** multiple myeloma is most common cause of renal and cardiac amyloidosis (this is all over the NBMEs); any renal/cardiac Dx in MM, answer = renal or cardiac amyloidosis.
- The IgG light chains show up in the urine → Bence-Jones proteinuria.
- The IgG light chains cause RBCs to stick together → Rouleaux formation + ↑↑ ESR (the Rouleaux are heavy, so the RBCs deposit at a faster rate).
- The neoplastic plasma cells can cause ↑ cytokine activity at bone → causes lytic lesions (e.g., "Pepperpot skull") + back pain (spinal lytic lesions) → lysis of bone causes hypercalcemia.



"Pepper pot skull"

- Smear in multiple myeloma will show plasma cells (blue cells below) with "clockface chromatin," which is the appearance ascribed to the nuclei (purple below).

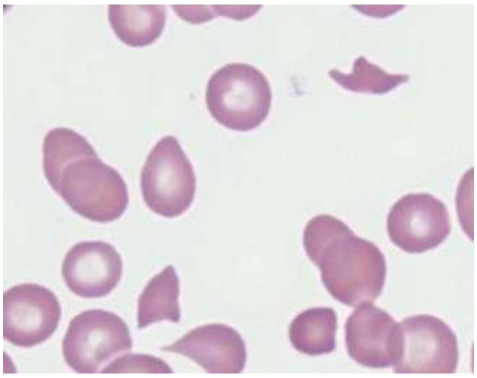


- "What are the highest yield points for Waldenstrom macroglobulinemia?"

    - Cancer of plasma**cytoid** cells → "oid" means looks like but ain't. So the cells look like plasma cells, but they ain't plasma cells. For instance,
        - Fibrinoid necrosis in polyarteritis nodosa → necrosis looks like fibrin, but it's not.
        - Patients with MEN2B have Marfanoid body habitus → they look like they have Marfan syndrome, but they don't.
    - Hyperviscosity syndrome → can present as headache, blurry vision, pain in the tips of the fingers, and/or Raynaud phenomenon.
        - Correct, Raynaud occurs in things other than CREST syndrome. And these hyperviscosity Sx are also seen in polycythemia.
    - There is an **IgM** M-protein spike; in contrast, MM has an IgG M-protein spike; once again, the M in M-protein spike has zero to do with IgM; it's just what we call the immunoglobulin spike. Students fuck this up a lot, which is why I have to be extra redundant explaining this.
    - Unlike MM, there is no hypercalcemia; there is no Bence-Jones proteinuria; ESR need not be elevated.
- 82F + back pain + hypercalcemia + high ESR; Dx? → multiple myeloma.
- 82F + back pain + hypercalcemia + Hx of breast cancer 15 years ago; Dx? → metastatic malignancy.
- 64M + back pain + calcium normal + high BUN & Cr + low serum Na + high serum K + low serum bicarb + x-ray shows lytic lesions of spine + epistaxis + low Hb + high MCV; Dx? → multiple myeloma causing renal amyloidosis, resulting in uremic platelet dysfunction and renal tubular acidosis type IV (insensitivity to aldosterone, or hyporeninemic hypoaldosteronism → Addisonian biochemistry but the problem **is due to the kidney, not the adrenal glands);** student says, "Wait, but why is calcium normal?" → Great question; probably because renal failure causes low Ca, but MM causes high Ca, so the patient could theoretically be normocalcemic in MM if he/she also has renal failure (one NBME Q has normocalcemia in MM with renal failure); epistaxis due to uremic platelet dysfunction; low Hb due to nosebleeds; MCV can be elevated in multiple myeloma.
- 50F + serum protein electrophoresis shows M-spike + bone marrow biopsy shows <10% plasma cells; Dx? → answer = monoclonal gammopathy of undetermined significance (MGUS); 1-2% chance per year of progressing to multiple myeloma.

- 28F + just gave birth + two minutes after separation of placenta gets shortness of breath and tachycardia + bleeding from IV + catheter sites; Dx? → answer = disseminated intravascular coagulation (DIC) secondary to amniotic fluid embolism.
- "Can you explain DIC?" → intractable clotting cascade activation in the setting of manifold triggers such as trauma, sepsis, amniotic fluid embolism, Tx of AML with release of Auer rods into circulation → USMLE wants you to know the arrows in DIC **(high-yield):**
  - ↑ BT, PT, aPTT, D-dimer, plasmin activity; ↓ fibrinogen, platelets, clotting factors.
  - Fibrinogen is converted to fibrin, which is why it decreases.
  - Plasmin breaks down fibrin, so more fibrin means more plasmin is upregulated in an attempt to dissolve the excess fibrin production.
  - D-dimer = fibrin degradation products; since more fibrin is being broken down, D-dimer ↑.
  - "Bleeding from catheter/IV sites" is 9 times out of 10 synonymous with DIC. However on one 2CK-level surgery NBME Q, dilutional thrombocytopenia secondary to ↑ blood transfusions presents the same.
- 40F + has surgery + requires 22 packs of RBCs transfused during the surgery + afterwards has bleeding from catheter/IV sites + DIC not listed as an answer; Dx? → answer = thrombocytopenia; student says, "Wtf? I thought that description is DIC." → packed RBCs don't contain platelets → ↑↑ acute transfusions → dilutional thrombocytopenia + can present with DIC-like picture in vignette. Weird I know, but you learn something new every day.
- 32F + SLE + DVT + PTT of 60 seconds; Dx? → answer = antiphospholipid syndrome (APLS) → APLS is the answer when the patient has thromboses despite a paradoxically high aPTT (you'd normally expect bleeding diathesis with high aPTT). This is because phospholipid is necessary for proper functioning of the clotting cascade, so the *in vitro* PTT test doesn't work as well (therefore ↑), but the antibodies cause clotting factor activation *in vivo.* APLS has numerous causes: you can have APL antibodies in the setting of SLE (we just happen to call these Abs lupus anticoagulant), but we can also have Abs against beta-2-microglobulin and cardiolipin, unrelated to SLE. Patients with APLS can sometimes have a false-positive VDRL screening test for syphilis.

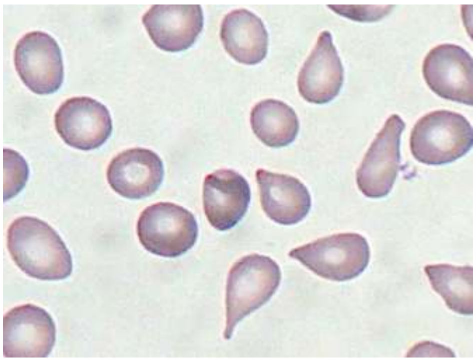
- Important coagulopathies? → FVL, prothrombin gene mutation, antithrombin deficiency, protein C/S deficiency.
- 65M + warfarin-induced skin necrosis; patient most likely has what? → answer = protein C deficiency → short half-life of protein C, which is depleted by warfarin, leads to ↑↑ hypercoagulable state in someone innately deficient.
- 23M + thrombosis + no mention of ↑ PTT; Dx (FVL, prothrombin mutation, protein C/S deficiency are not listed)? → answer = antithrombin deficiency; wrong answer is antiphospholipid syndrome (presumably for APLS the vignette will mention ↑ PTT if male patient, or recurrent miscarriage in female).
- 13M + bloody diarrhea + afebrile + low platelets + low Hb + red urine + smear is shown; Dx?



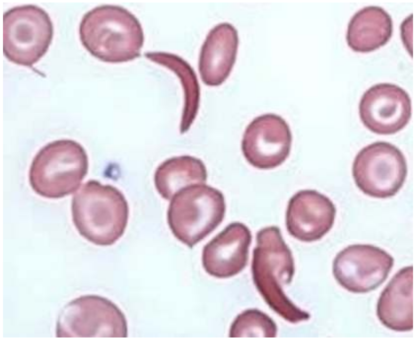
  - Answer = hemolytic uremic syndrome (HUS); presents with triad of:
  - 1) thrombocytopenia;
  - 2) hemolytic anemia with schistocytes (above smear); and
  - 3) renal insufficiency with or without hematuria.
- Mechanism of HUS? → E. coli (EHEC O157:H7) and Shigella both secrete toxins (Shiga-like toxin and Shiga toxin, respectively) that cause inflammation of renal microvasculature → ADAMTS13 protein inactivation → failure of cleavage of vWF multimers → platelet adherence to vascular endothelium cannot be as readily reversed → platelet aggregations protrude into vascular lumen causing shearing of RBCs flying past → fragmentation of RBCs (schistocytes, aka helmet cells).
- 34F + epistaxis + platelet count 80,000 + low Hb + muscle strength 3/5 on left side of body + temperature 101.2F; Dx? → answer = thrombotic thrombocytopenic purpura (TTP) → presents with

- "Can you explain TTP? I always mix that up with ITP." → TTP is thrombotic thrombocytopenic purpura; caused by antibodies against, or a mutation in, a protein called ADAMTS13, which is a metalloproteinase that breaks down von Willebrand factor multimers. The presentation will classically be a pentad of 1) thrombocytopenia; 2) schistocytosis; 3) renal insufficiency; 4) fever; 5) neurologic signs. The combination of thrombocytopenia + schistocytosis = microangiopathic hemolytic anemia (MAHA).
- 34F + epistaxis + platelet count 80,000 + low Hb + muscle strength 3/5 on left side of body + temperature 101.2F; Dx? → answer = TTP; at first you're like, "Is this a stroke? What's going on here?" This is how it presents on the NBME. The low hemoglobin is due to the fragmentation of the RBCs leading to schistocytosis.
- "What do you mean fragmentation of RBCs leading to schistocytosis?" → schistocytes are caused by the fragmentation/shearing of RBCs intravascularly. TTP is one of the common causes. The others are hemolytic uremic syndrome (HUS), DIC, HELLP syndrome in pregnancy, and mechanical hemolysis (prosthetic heart valves).
- "Can you explain HUS? And I confuse it sometimes with TTP." → therefore the combination of thrombocytopenia + schistocytosis (often seen in a Q as a combo of low platelets + low Hb + normal WBC count) should set off alarm bells for HUS and TTP. Because this process occurs in microvasculature, this is why the combo of thrombocytopenia and schistocytosis is referred to as microangiopathic hemolytic anemia (MAHA). Other HY points:
    - Whereas TTP presents with a pentad of 1) thrombocytopenia; 2) schistocytosis; 3) renal insufficiency; 4) fever; 5) neurologic signs, HUS presents with just the first three. Therefore:
    - **TTP = HUS + fever and neurologic signs.**
    - The mechanisms are different, as discussed above, but presentation-wise, that's an easy way to remember it.
    - HUS will classically be pediatric + bloody diarrhea (EHEC or Shigella); TTP will usually be an adult who presents with fever and stroke-like Sx + who also has the triad you'd expect to see in HUS. The tricky thing about TTP is that episodes can be triggered by various etiologies, such as viral infection, but the vignette will not give bloody diarrhea.
    - Tx for HUS is supportive with intravenous fluids; Tx for TTP is plasmapheresis.

- 63M + fullness in LUQ + low Hb + high uric acid + bone marrow aspiration shows dry tap; Dx? → answer = myelofibrosis → "massive splenomegaly" often seen in NBME (fullness of LUQ); Hb can be low from myelophthisic anemia (crowding of bone marrow leading to anemia); myelofibrosis due to JAK2 mutation (same as PCV). Student says, "Wait but why high uric acid? Isn't that gout?" → can be seen sometimes with increased cell turnover, including RBCs (precursors have nucleic acid).
- 54F + massive splenomegaly + smear is shown; Dx?



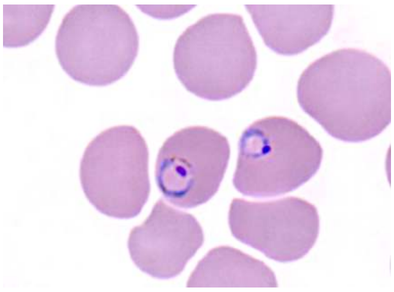
- Answer = myelofibrosis → dacrocytes = teardrop-shaped RBCs → HY for myelofibrosis; key terms are: "massive splenomegaly," "dry tap," "teardrop RBCs").
- What does red pulp vs white pulp of spleen do? → red pulp is where senescent RBCs are phagocytosed; white pulp contains ~50% of the body's reservoir of macrophages.
- 3F + African-American + painful hands + HR120 + 2/6 mid-systolic murmur; Dx? → answer = sickle cell crisis; dactylitis (inflammation of the fingers) is one of the most common presentations, especially in pediatrics; benign flow/functional murmurs can be seen with high HR, especially in peds.
- Inheritance pattern of sickle cell? → answer = **autosomal recessive (HY)**; carrier status (one mutation) is referred to as sickle cell trait, which is less severe than sickle cell anemia (two mutations).
- Mutation in sickle cell? → answer = glutamic acid → valine on the beta-chain.
- How to Dx sickle cell? → answer = hemoglobin electrophoresis → glutamic acid is negatively charged; valine is neutral; this means HbS is more non-polar → does not migrate as far on Hb electrophoresis (gel goes from - → +) because it is less attracted to the + end of the gel.
- When does sickling notably occur in sickle cell? → dehydration + increased acidity.



Sickle cells

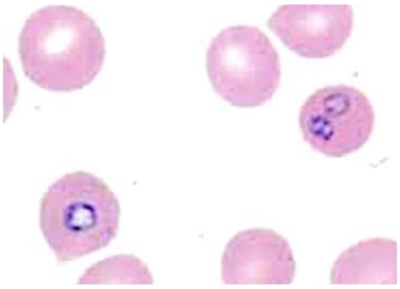
- 4M + sickle cell + Q asks which of the following best describes the molecular basis for sickling in this patient → answer = "gain of stabilizing hydrophobic interactions in the deoxygenated form of hemoglobin S" → valine is more hydrophobic than glutamic acid.
- 5F + sickle cell; Q asks mechanism for sickling; answer = beta chain slips into a complimentary hydrophobic pocket on the alpha chain.
- 3F + takes penicillin prophylaxis; why? → decreases Strep pneumo infections; penicillin prophylaxis indicated until age 5, in addition to PCV13 and PPSV23 Strep pneumo vaccines.
- Why autosplenectomy in sickle cell + what vaccines are needed? → sickling in red pulp leads to microinfarcts → any sickle cell patient needs vaccines for *S. pneumo, H. influenzae type B,* and *Neisseria meningitidis.*
- Why are those vaccines needed? → with splenectomy (or autosplenectomy), increased risk of infection with encapsulated organisms, which require opsonization and phagocytosis for clearance → spleen white pulp is important site of phagocytosis (tangential: IgG and C3b are immune system's main opsonins).
- One-year-old girl + missed dose of penicillin prophylaxis a few days ago + now has fever of 103F + HR 100, RR 22, low BP; what antibiotic do we give? → answer = cefotaxime; penicillin and ceftriaxone are wrong answers; community-acquired sepsis (patient need not have sickle cell) often treated with third-generation cephalosporin; give cefotaxime < age 6; give ceftriaxone > age 6.
- 8M + sepsis; Abx Tx? → answer = ceftriaxone.
- 5F + sickle cell + red urine; Dx? → answer = renal papillary necrosis; most common cause of nephritic syndrome in sickle cell.

- 5F + sickle cell + renal problem + no blood in urine; Dx? → answer = focal segmental glomerulosclerosis (FSGS); most common cause of nephrotic syndrome in sickle cell.
- Osteomyelitis organism in sickle cell? → answer = *Salmonella.*
- 3F + sickle cell + foot pain for three weeks + fever of 103F + high WBCs; Dx? → answer = osteomyelitis; acute sickling crisis is wrong answer; difficult Q, as low-grade fever, leukocytosis, and pain over many weeks can be seen in sickling crisis, however fevers >101F suggest infection; osteomyelitis may present with pain over many weeks.
- 16F + sickle cell; what is this patient at increased risk of? → answer = cholelithiasis; USMLE loves this one → ↑ RBC turnover → ↑ unconjugated bilirubin production → ↑ risk of pigmented gall stones (calcium bilirubinate).
- Tx for sickle cell + what's the mechanism? → answer = hydroxyurea; mechanism = ↑ production of fetal hemoglobin (HbF); hydroxyurea is a ribonucleotide reductase inhibitor → inhibits pyrimidine synthesis.
- "Why do those with sickle cell trait and anemia have ↑ resistance to malaria?" → RBC lifespan is reduced due to ↑ RBC turnover, so intra-erythrocytic lifecycle of the protozoan is disrupted.
- How to Dx malaria? → answer = thick and thin blood smears.
- Which type of malaria is the "worst?" → answer = *Plasmodium falciparum* because it causes cerebral malaria (microthrombosis and hemorrhage).
- Malaria prophylaxis? → chloroquine, mefloquine.
- MOA of chloroquine/mefloquine? → inhibit *Plasmodium* heme polymerase.
- 20F + went to Africa + took chloroquine prophylaxis + got malaria anyway; why? → answer = chloroquine resistance; wrong answer is medical noncompliance. Chloroquine resistance is HY.
- 20F + has malaria + treated appropriately + several weeks later has a resurgence of the malaria; why? → answer = "presence of extra-erythrocytic form of organism" → *P. vivax* and *ovale* cause hypnozoites, which are a latent intrahepatic form of the disease.
- 20F + has *P. vivax* + treated with primaquine; why? → answer = "primaquine kills hypnozoites."
- 20F + comes back from Africa with hemolytic disease + smear is shown; Dx?



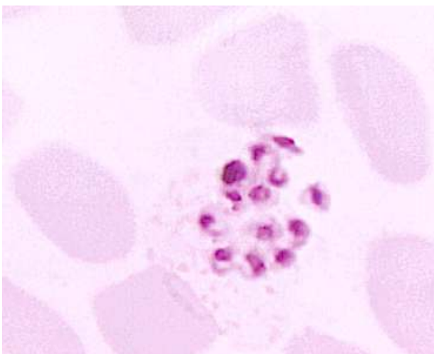
 - Answer = malaria; smear shows ring form of malaria.

- 20F + lives in Connecticut + has a hemolytic disease; smear is shown; Dx?



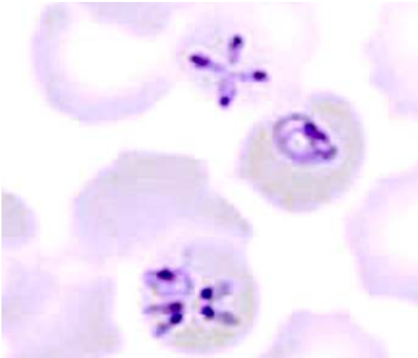
 - Answer = Babesia; smear shows ring form.
 - Babesia causes a hemolytic disease similar to malaria **but the patient will not have left the USA.** This type of Q can be tricky because both malaria and Babesia can have a similar-appearing ring form. Bottom line:
   - Hemolytic disease + ring form on smear + patient went to Africa, South America, or Asia; answer = malaria.
   - Hemolytic disease + ring form on smear + patient never left the United States; answer = Babesia, not malaria.

- 22F + comes back from Africa with hemolytic disease + smear is shown; Dx?



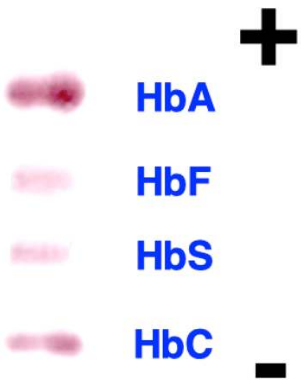
 - Answer = malaria; smear shows schizont form.

- 22F + lives in Connecticut + has a hemolytic disease; smear is shown; Dx?



- Answer = Babesia; smear shows characteristic Maltese cross.

- "What is HbC?" → just another type of Hb disorder where glutamic acid → lysine (not valine); lysine is positively charged, so it migrates the least far on Hb electrophoresis, as it is most attracted to the – charge at the origin (gel goes from - → +).



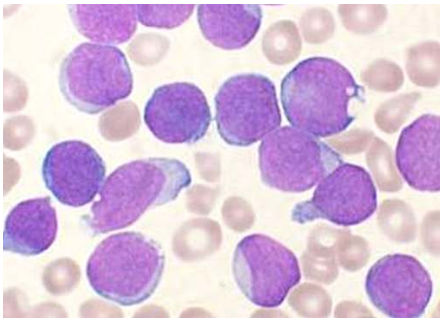
- What variables shift the Hb-O2 curve to the right? → answer = ↑ temperature, ↑ CO2, ↑ H^+ (i.e., ↑ acidity), ↓ $HCO3^-$, ↑ 2,3-BPG.
- What does a Hb-O2 curve shift to the right mean? → answer = increased unloading of oxygen at tissues.
- Does anything shift the curve to the left? → notably fetal hemoglobin (HbF) has left-shifted curve.
- Why is HbF ($\alpha 2\chi 2$) left-shifted compared to adult hemoglobin (HbA1; $\alpha 2\beta 2$)? → The beta chain on adult hemoglobin has a charged histidine that forms an ionic bond with 2,3-BPG; this histidine is replaced with an uncharged serine on the gamma chain of HbF that does not bind 2,3-BPG.
- "Why can deoxygenated blood carry more CO2 for a given pCO2 than oxygenated blood?" → answer = "deoxyhemoglobin is a better buffer of hydrogen ions than oxyhemoglobin." Student immediately says wtf? → At peripheral tissues, CO2 production is ↑; this diffuses into the RBC, combines with water to make H2CO3, which then equilibrates to bicarb and a proton. **The proton hops onto the**

deoxygenated hemoglobin – i.e., the deoxyhemoglobin acts as a buffer for protons in the blood. At the same time, the bicarb moves out of the RBC into the plasma, and chloride moves into the RBC to balance charge. This is referred to as chloride shift. Key points:

- At the tissues, CO_2 moves into RBCs.
- $CO_2 + H_2O \leftrightarrow H_2CO_3 \leftrightarrow HCO_3^- + H^+$.
- HCO_3^- leaves the RBC. Cl^- moves into RBC to balance charge. H^+ hops onto deoxy-Hb.
- Therefore most CO_2 in the blood is carried as **bicarbonate in the plasma.** Students tend to erroneously pick "bicarb in the RBC" because it sounds weird so they think it's right. But the answer is bicarb in plasma.
- At the lungs, this process reverses, where H^+ hops off the Hb, Cl^- leaves the RBC, HCO_3^- enters the RBC. $HCO_3^- + H^+ \leftrightarrow H_2CO_3 \leftrightarrow CO_2 + H_2O$. CO_2 then leaves the RBC and is exhaled.

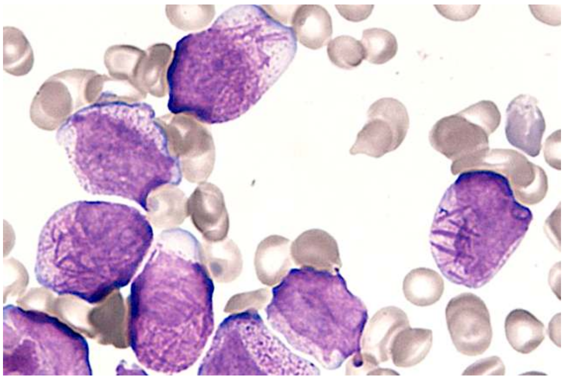
- 43M + works as a butcher + recent fatigue + low O_2 sats; Dx? → answer = methemoglobinemia; can be caused by nitrates used to preserve meats.
- Mechanism of methemoglobinemia? → Fe on hemoglobin is normally 2^+ charge state (ferrous); it becomes oxidized to $Fe3^+$ (ferric), which does not bind O_2 as well → Hb becomes desaturated.
- 28F + goes hiking and drinks mountain water + "brown blood"; Dx? → answer = methemoglobinemia caused by nitrates/nitrates (yes, both can cause it) found in mountain/river water; "brown blood" is seen in methemoglobinemia; in contrast "cherry red blood" (or lips) is seen in CO poisoning.
- 3F + "brown blood"; mechanism for her disease? → answer = "congenital methemoglobinemia caused by deficiency of cytochrome B5 reductase."
- Tx for methemoglobinemia? → answer = IV methylene blue + vitamin C.
- Pulse oximetry finding in methemoglobinemia? → answer = low (80s%).
- 34M + fatigue + moved into a new house in winter with an old ventilator; Tx? → answer = hyperbaric oxygen for CO poisoning.
- 34M + light-headedness + was hanging out on moored boat while the engine was running; Dx? → CO poisoning.
- Mechanism for CO poisoning? → CO has ↑↑ higher affinity for Hb than O_2, so maximal O_2 binding to Hb is impaired.

- Pulse oximetry finding in CO poisoning? → answer = **pulse oximetry is normal is CO poisoning** → standard pulse oximeters can only read the gas bound to Hb, period; they can't distinguish whether it's O2 or CO; specialized CO oximeters do exist that can distinguish.
- What is normal pulse oximetry finding? → 98-100% saturation. However values as low as 94% can be considered acceptable.
- 34M + new-onset fatigue + lives in house in winter with new ventilator + also just purchased second-hand refrigerator + pulse oximetry 94%; Q asks "What in this patient's house is causing his condition?" → answer = ventilator → Dx is CO poisoning; the mention of the refrigerator in the above Q is a distractor (for those thinking cyanide toxicity, the association is not classic); student says, "Wait, I thought you just said pulse oximetry is normal in CO poisoning. Isn't normal 98-100%?" → Yes, but Qs have been known to have Hb saturation as low as 94% in CO poisoning.
- Tx for CO poisoning? → answer = hyperbaric oxygen.
- How does CO affect the Hb-O2 curve? → down-shifts it.
- 34M + housefire + confusion + burned upholstery + O2 sats normal; Dx? → answer = cyanide toxicity; classically caused by inhaling fumes from burned upholstery.
- 34M + BP of 250/130 + confusion; Dx? → hypertensive encephalopathy. You say, "What? How does that relate to heme/onc?" Stay with me here:
- 34M + BP of 250/130 + no confusion; sodium nitroprusside is administered *then* develops confusion; Dx? → answer = cyanide toxicity, not hypertensive encephalopathy; CN toxicity can be caused by sodium nitroprusside. Dumb trick, but on USMLE.
- Mechanism for CN toxicity? → binds to cytochrome oxidase and "inhibits transfer of electrons to molecular oxygen"; that is: it interferes with the electron-transport chain.
- Tx for cyanide toxicity? → answer = amyl nitrite (answer on NBME) → the nitrites cause methemoglobinemia; cyanide binds readily to Fe^{3+}, forming cyanmethemoglobin, thereby releasing the CN from cytochrome oxidase. Sodium thiosulfate and hydroxocobalamin are also used.
- 4F + leukocytes 78,000/uL (87% lymphocytes) + low Hb + low platelets + lymphadenopathy; Dx? → answer = acute lymphoblastic leukemia (ALL) → pretty much always the answer for leukemia in peds → normal leukocyte count is 4-11,000/uL → high leukocytes in peds with lymphocyte-predominant spread should scream ALL. Leukemia is usually **B cell.**



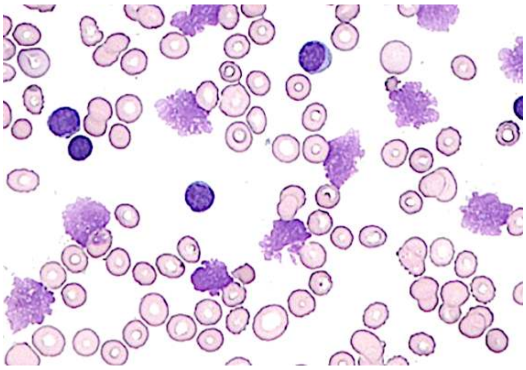
ALL; lymphoblasts appear smooth and relatively uniform.

- 6M + WBCs 56,000 (mostly lymphocytes) + flushing of the face; Dx? → T cell ALL (TALL) → if SVC-like syndrome is seen due to thymic lesion, answer is T cell, not B cell, variant.
- 3F + one-month Hx of fatigue + WBCs 3500 + low Hb + low platelets + lymphadenopathy; next best step in Dx? → answer = bone marrow aspiration → Dx = ALL. Student says, "I thought you just said though that WBCs would be high." → can rarely have normal or low leukocyte counts; one NBME Q has vignette resembling aplastic anemia + lymphadenopathy → answer is bone marrow aspiration, which is confirmatory for ALL. The non-acute presentation suggests leukemia over viral-induced aplastic anemia.
- 3M + trisomy 21 + pancytopenia + examination of bone marrow will show what? → answer = excess lymphoblasts → increased risk of ALL in Down syndrome (and AML type VII, but in peds the answer is almost always ALL).
- 20M + lymphadenopathy + low platelets + smear is shown; Dx?



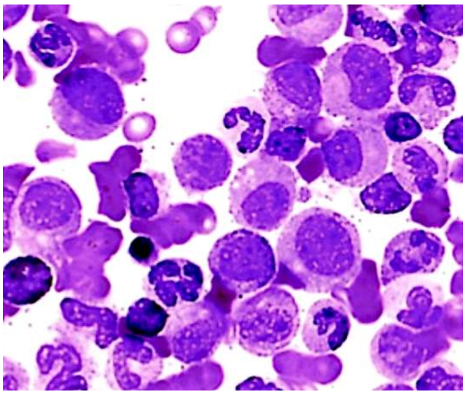
  - Answer = acute myelogenous leukemia (AML) → image of Auer rods is exceedingly HY for the Step; Auer rods are composed of myeloperoxidase, which is a blue-green heme-containing pigment; Tx of AML leads to lysis of cells → Auer rods released into circulation and precipitate DIC.

- Tx of the leukemia + increased serum uric acid levels; Dx? → answer = tumor lysis syndrome; need to know the arrows → answer = ↑ potassium; ↓ bicarb; ↓ calcium; ↑ phosphate; ↓ CO2; variable sodium; ↑ uric acid. Xanthine oxidase inhibitors can help prevent (i.e., allopurinol, febuxostat); do not give these agents if 6-mercaptopurine or azathioprine are being used in Mx (require XO for breakdown).
- 56M + ↑↑ WBCs + t(15;17); Dx + Tx? → answer = acute promyelocytic leukemia (AML type M3); Tx is all-trans retinoic acid (vitamin A).
- 82F + fever 103F + gram (+) diplococci on sputum sample + WBCs 82,000 (87% lymphocytes); Dx + next best step in management? → answer = chronic lymphocytic leukemia (CLL); next best step → NBME answer = “quantitative immunoglobulin assay” → apparently ordered for those with chronic infections when IgG is low (can be seen in leukemia). HY points:
    - Whenever I ask students about this type of vignette, they’ll always say it’s Strep pneumo causing pneumonia. But if this were the Dx in isolation, WBC count should only be about 11-20,000 at most; if the patient is septic, maybe upwards of 25-30,000. So when you see WBCs >30,000, you really need to say, “Oh shit ok, that’s leukemia as the underlying Dx. And this patient just happens to have an infection due to immunodeficiency.”
    - In addition, Strep pneumo causes an extracellular bacterial infection, so the shift should be toward **neutrophils (normal range ~55-60%,** where bacterial shift would be ~65-90%), so if the shift is toward lymphocytes, that should scream ALL or CLL. ALL would be the answer for peds, however, not adults.
    - Important you’re tangentially aware that pertussis can cause WBCs >30,000 with a lymphocyte shift; resembles ALL in peds; always a wtf finding when you first learn of it; but just say: “Ok, super high WBCs with lymphocyte shift → ALL, CLL, or pertussis.”
- 55M + high WBCs + Coombs test positive + smear is shown; Dx?



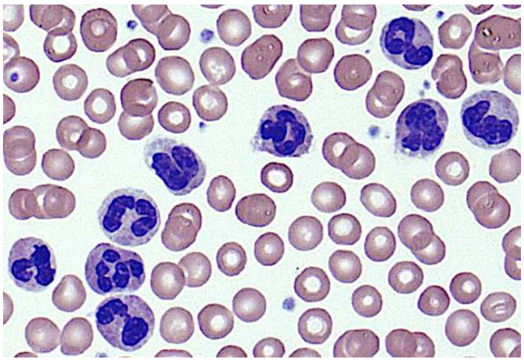
- Answer = CLL; smear shows smudge cells; CLL sometimes associated with autoimmune hemolytic anemia (usually warm), hence a (+) Coombs test may be seen.
- "Warm? What?" → Relax. Warm vs cold autoimmune hemolytic anemia (AIHA) means you've got either IgG (warm) or Ig**M** (cold; **Mmm** ice cream) against RBCs.
- Cold AIHA (aka cold agglutinin disease) is seen sometimes with mycoplasma or CMV infection, where the patient can have IgM antibodies against RBCs and a hemolytic anemia (e.g., CMV infection + low Hb).
- Positive Coombs test = the patient has IgM or IgG antibodies against RBCs → **whatever RBC issue the patient has, it's due to antibodies.**

- "Wait, can you explain Coombs test real quick?" → direct vs indirect types; direct Coombs is taking the patient's RBCs and seeing if they agglutinate *in vitro* using various laboratory antibodies; if the RBCs agglutinate, this means there were antibodies attached to their surface, and the patient did in fact have antibody-mediated hemolytic anemia; indirect Coombs is taking the patient's plasma and seeing if it induces laboratory RBCs to agglutinate; if agglutination occurs, then the patient had antibodies in his/her plasma and we know that he/she did in fact have antibody-mediated hemolysis.
- 61F + high WBCs + CD5 and CD23 positivity + positive Coombs test; Dx? → answer = CLL; leukemic cells in CLL can be CD5 and/or CD23 positive (on retired Step 1 NBME).
- 44F + WBCs 14,500 + metamyelocytes and myelocytes seen on FBC + smear is shown; Dx?



  - Answer = CML; smear shows "motley mix" of many different types of cells (HY image).

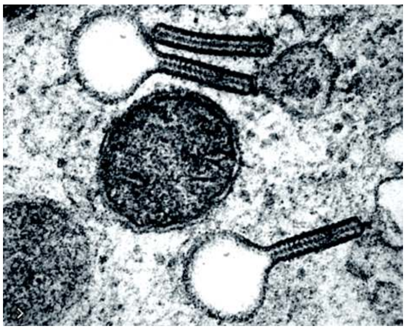
- 44F + WBCs 32,000 + metamyelocytes and myelocytes seen on FBC + urinalysis shows nitrites and leukocyte esterase + smear is shown; Dx?



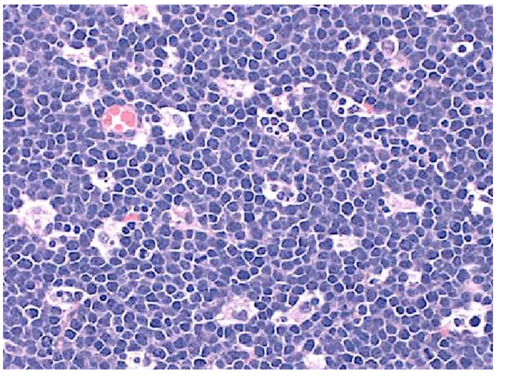
  - Answer = leukemoid reaction; smear shows neutrophilia consistent with infection (UTI in this case); answer can also be written on NBME as "reactive granulocytosis."
  - Leukemoid reaction = increased release of leukocytes from bone marrow reserve pool. WBC count will be >30k. There is NBME Q where it's 32k.

- 44F + WBCs 14,500 + metamyelocytes and myelocytes seen on FBC + decreased leukocyte ALP; Dx? → answer = chronic **myelo**genous leukemia (CML); meta**myelo**cytes and **myelo**cytes are extremely HY for CML; leukocyte ALP is also decreased.
- 44F + WBCs 32,000 + metamyelocytes and myelocytes seen on FBC + increased leukocyte ALP; Dx? → answer = leukemoid reaction (inflammatory process; usually infection); the other HY condition that, like CML, can present with metamyelocytes and myelocytes, however leukocyte ALP is increased.
- 44F + WBC 180,000 (50% neutrophils) + t(9;22); Dx? → answer = CML → Philadelphia chromosome → t(9;22) bcr/abl → oncogenic tyrosine kinase.
- Tx of CML? → answer = imatinib.

- Important side-effect of imatinib? → answer = fluid retention (edema); tangentially, apart from imatinib, know that dihydropyridine calcium channel blockers (nifedipine, amlodipine, etc.) also cause fluid retention/edema.
- 54M + lytic bone lesions + electron microscopy is shown with cells that are CD1a positive; Dx?



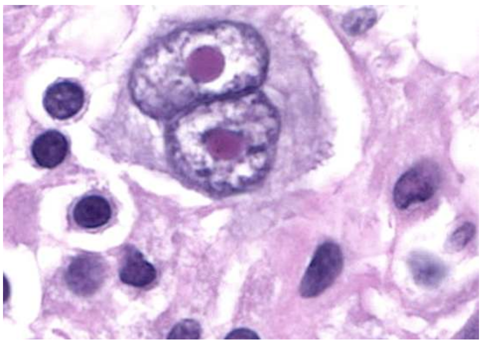
  - Answer = Langerhan cell histiocytosis; characteristic tennis racquet-shaped cells are called Burbeck granules.
- 17M + fever + tonsillar exudates + cervical lymphadenopathy + confluent ulcerations seen in posterior oropharynx; this pathogen can also cause what? → answer = hemolytic anemia; patient has mononucleosis; 90% of the time, it's caused by EBV; but 10% is CMV; linear (confluent) ulcers = CMV; CMV is a known cause of cold AIHA.
- 9M + African-American + abdominal mass growing left of umbilicus + fever + night sweats + weight loss + cytogenetic analysis reveals t(8;14); Dx? → answer = Burkitt lymphoma; students says, "Wait, I thought it was supposed to be a jaw lesion." → can be intra-abdominal or jaw.
- 9M + African-American + unilateral jaw swelling + t(2;8) + biopsy of lesion is shown; Dx?



  - Answer = Burkitt; image is classic "starry sky" appearance; blue cells are lymphocytes; clear cells are macrophages.

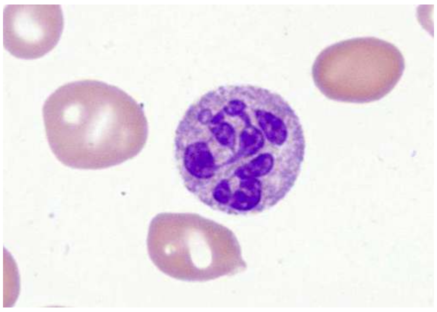
- NBME Q for Step 1 points to a macrophage on the starry sky histo and asks what cellular process is occurring; answer = apoptosis → macrophages are called tingible (not tangible) body macrophages → contain lots of phagocytosed cells at various stages of apoptosis.
- USMLE Step 1 assesses t(2;8), t(8;14), and (8;22) for Burkitt; resources tend to only focus on t(8;14); easy way to remember: you can see that chromosome 8 is involved in all of them.

- 55M + fever + night sweats + weight loss + fluid wave in abdomen + abdominal paracentesis yields 300mL of milky fluid; Dx? → chylothorax secondary to intra-abdominal Burkitt lymphoma.
- Gene involved in Burkitt + what is it? → c-myc → transcription factor.
- 26M + waxing and waning neck mass over one-year period + t(14;18); Dx? → answer = follicular lymphoma; may present with waxing/waning disease.
- Gene involved in follicular lymphoma + what is it? → answer = bcl-2 → anti-apoptotic molecule.
- 41F + lymphoma + t(11;14); Dx? → answer = mantle cell lymphoma.
- 9F + WBC 3,500 + smear shows WBCs with cytoplasmic projections that stain positive for acid-resistant acid phosphatase; Dx? → answer = hairy cell leukemia; can have low WBC count.
- 72M + acute-onset highly aggressive B cell lymphoma; Dx? → diffuse-large B cell lymphoma (DLBCL); most common non-Hodgkin lymphoma (NHL) in adults; highly aggressive.
- "What does non-Hodgkin vs Hodgkin mean?" → Hodgkin is a type of lymphoma that has pathognomonic Reed-Sternberg cells on lymph node biopsy (owl-eye appearance); RS cells are CD15 and CD30 positive; some resources say NHL has B-symptoms (fever, night sweats, weight loss) whereas in Hodgkin they are more rare, but on the USMLE, B-symptoms can occur in either; EBV can also cause either; Hodgkin may present with contiguous spread, where affected areas are in close proximity, whereas NHL can spread more haphazardly. NHL just refers to any lymphoma that doesn't have Reed-Sternberg cells.



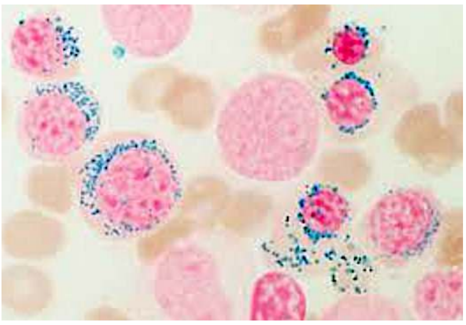
Reed-Sternberg cells

- "Do I need to know the different types of Hodgkin lymphoma?" → highest yield points:
    - Nodular sclerosing variant is more common in women.
    - More Reed-Sternberg cells = poorer prognosis.
        - Leukocyte-rich Hodgkin = high lymphocytes + low RS cells (better prognosis).
        - Leukocyte-deplete Hodgkin = low lymphocytes + high RS cells (poor prognosis).
- 44M + Hx of Hodgkin disease + nephrotic syndrome; what's the renal Dx? → answer = minimal change disease (MCD); student says, "Wtf? But I thought that's pediatrics." → 9 times out of 10, yes, it follows viral infection in a kid, but it is also associated with Hodgkin lymphoma in adults.
- 16F + painless lateral neck mass + a mediastinal mass; Dx? → answer = Hodgkin disease
- 16F + painless lateral neck mass + hepatomegaly; Dx? → answer= Hodgkin disease → classic vignette presents as painless lateral neck mass (doesn't wax and wane like follicular NHL) + either a mediastinal mass or hepatomegaly; the mediastinal mass is not a thymoma; it's mediastinal lymphadenopathy.
- 49M + jaundice + high ALP + pancreatic enzymes normal + weight loss + painful erythematous areas on arms and legs; Dx? → pancreatic head adenocarcinoma causing migratory thrombophlebitis (Trousseau sign of malignancy).
- 50M + hepatitis C + purpura on arms and legs + joint pain + low complement C4; Dx? → answer = cryoglobulinemia.
- "Wtf is a cryoglobulin?" → cryoglobulins are immunoglobulins that precipitate at "cold" temperatures (i.e., <37C); cryoglobulinemia can be caused by malignancy as well as chronic infections (HepC, HIV). C4 can be decreased due to activation of the complement cascade, notably C4. Do not confuse cryoglobulinemia with cold agglutinin disease, which is aka cold autoimmune hemolytic anemia.
- 22M + vegetarian/vegan + ↑ MCV + smear is shown; Dx?



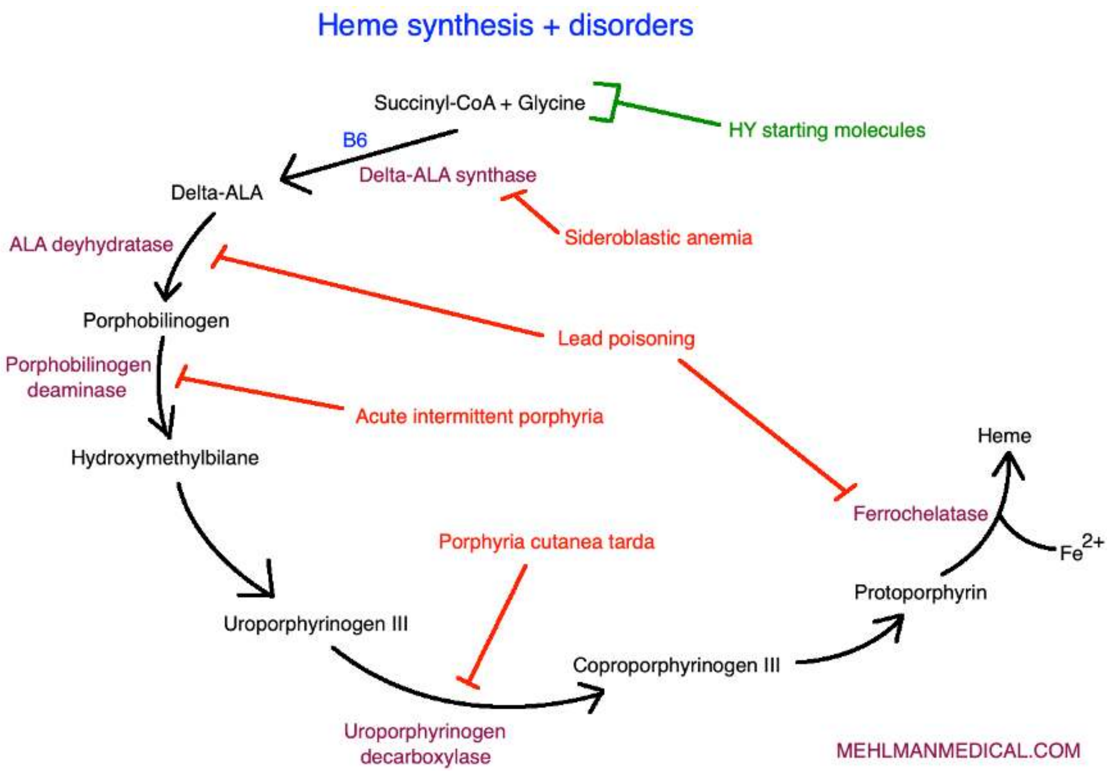
- Answer = dietary B12 deficiency; smear shows hypersegmented neutrophil.

- "What is a hypersegmented neutrophil?" → seen in folate (B9) or B12 deficiency; ↓ DNA synthesis results in neutrophils with >3 segments to the nuclei. Essentially if you see these on a smear, right away you should be thinking B9 or B12 deficiency; patient will also have ↑ MCV.
- 22M + vitiligo + vegetarian + ↑ MCV + hypersegmented neutrophils; Dx? → answer = pernicious anemia causing B12 deficiency (one autoimmune disease → ↑ risk of others [polyglandular syndromes]).
- 22M + vitiligo + vegetarian + ↑ MCV + hypersegmented neutrophils; next best step? → answer = check serum antibodies against parietal cells and intrinsic factor.
- 27F + strict vegetarian diet for 5 years + ↓ Hb + ↓ Hct + ↓ WBCs + ↓ platelets + no other info given; Q asks, her nutrient deficiency significantly impairs which of the following cellular processes; answer = DNA synthesis (B12 deficiency); wrong answer is heme production (instead B6 deficiency).
- 22M + Hx of epilepsy + ↑ MCV + hypersegmented neutrophils; Dx? → B9 deficiency → anti-epileptic meds (i.e., valproic acid, phenytoin, carbamazepine) cause ↓ intestinal folate absorption.
- 44M + alcoholic + ↑ MCV + no hypersegmented neutrophils + serum methylmalonic acid and homocysteine levels normal + smear is shown; Dx?



- Answer = alcohol-induced sideroblastic anemia → alcohol can cause ↑ MCV. But since there's no impairment of DNA synthesis, there are no hypersegmented neutrophils; alcohol can merely disrupt the heme synthesis pathway, where MCV can sometimes ↑.
- Methylmalonic acid (or methylmalonyl-CoA) is ↑ in B12 deficiency; homocysteine is ↑ in both B9 and B12 deficiencies.

- "What is sideroblastic anemia?" → condition characterized by normal iron levels but merely the inability to incorporate the iron into heme → results in RBC precursors (nucleated erythroblasts) containing peri-nuclear iron-laden macrophages that stain blue with Prussian blue stain ("ringed sideroblasts"). The visible iron aggregates are known as Pappenheimer bodies.
- "Do I need to know the heme synthesis pathway?" → Annoying, but yes. The heme pathway disorders are all over the NBMEs for Step 1.

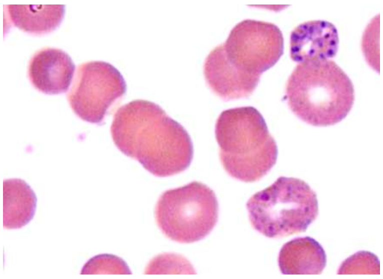


- Cause of sideroblastic anemia? → Usually X-linked recessive condition due to deficiency of δ-aminolevulinic acid synthase (δ-ALA), but can also be acquired, most commonly due to alcohol.
- Classic hematologic parameters in sideroblastic anemia? → ↑ serum iron + ↑ ferritin + ↑ transferrin saturation + ↓ Hb; MCV can be normal/↓ in XR form or ↑ MCV in acquired forms.
  - Bottom line: remember ↑ iron across the board + MCV can be variable.

- 43M + daily alcohol use + ↑ serum iron + ↑ ferritin + ↑ transferrin saturation + ↓ Hb; Dx? → answer = alcohol-induced sideroblastic anemia.
- 23F + abdominal pain + pink urine + ↑ urinary porphobilinogen + ↑ δ-ALA; Dx? → answer = acute intermittent porphyria → heme synthesis disorder caused by deficiency of porphobilinogen deaminase → classically associated with "Port wine-colored urine" (but vignettes can just say pink or red), abdominal pain, and ↑ urinary porphobilinogen; some patients can also have neurologic findings.
- 19F + weakness of legs + decreased reflexes + severe abdominal pain + persistent vomiting + Port-wine-colored urine; Q asks, urine studies are most likely to show ↑ what? → answer = porphobilogen; Dx is acute intermittent porphyria.
- Inheritance pattern of acute intermittent porphyria? → **autosomal dominant** (on Step 1 NBME).
- 44F + abdominal pain + no mention of urinary findings + paresthesias + alcohol seems to precipitate episodes; Dx? → answer = acute intermittent porphyria; can be exacerbated by alcohol; one 2CK-level neuro Q doesn't mention anything about pink/red urine but mentions neurologic findings.
- Tx for acute intermittent porphyria? → answer = glucose infusion (acutely ↓ heme synthesis); can also give hematin + heme arginate).
- 34F + recurrent episodes of blistering of face and arms over many years + ↑ serum ALT and AST + ↑ total serum porphyrin + ↑ urine uroporphyrin III; Dx? → answer = porphyria cutanea tarda → heme synthesis disorder caused by deficiency of uroporphyrinogen III decarboxylase → ↑ urinary uroporphyrin; causes "tea-colored urine" and photosensitivity (i.e., blistering).
- Tx for porphyria cutanea tarda? → answer = ↓ alcohol use (can precipitate Sx) + ↓ sun exposure.
- 40M + episodes of blistering from sun + ↑ urine uroporphyrin III; Q asks which compound serves as the precursor to uroporphyrin in this patient → answer = succinyl-CoA → required for the initiation of heme synthesis (glycine not listed but by all means also correct).
- 29M from Albania + positive PPD test + negative CXR + started on monotherapy for condition + develops paresthesias months later; which other finding might be seen in this patient? → answer = impairment of heme synthesis → patient being treated with isoniazid (INH) for latent TB → INH

causes vitamin B6 deficiency if not supplemented → presents as neuropathy and/or seizures. Vitamin B6 is needed for the first step of heme synthesis.

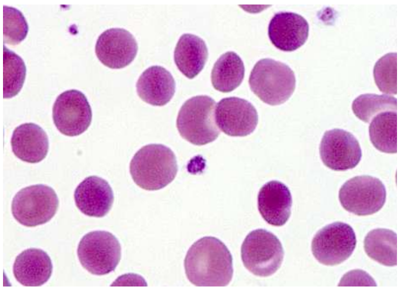
- 44M + hunter + recent cognitive decline + microcytic anemia + wrist drop + ↑ δ-ALA + ↑ RBC protoporphyrin; Dx? → answer = lead poisoning → inhibits ferrochelatase (causes ↑ RBC protoporphyrin) and δ-ALA dehydratase (↑ δ-ALA); lead poisoning classically causes microcytic anemia (HY finding, especially in adults; vignette is not always going to say kid eating paint chips in family's new house); and neuropathy (e.g., wrist drop, foot drop).
- 2M + cognitive decline after family moves into new house; what does the smear show?



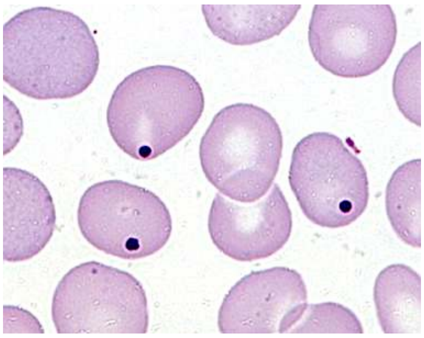
  - Answer = basophilic stippling (RNA precipitates) seen in lead poisoning.
- Tx for lead toxicity? → Tx not indicated unless serum levels >44 ng/dL (weirdly specific, but asked on 2CK/3); give dimercaprol or EDTA in adults; succimer is often used for children.
- How does iron toxicity present? → GI bleeding HY on USMLE; can also cause ↑ anion-gap metabolic acidosis (Iron and INH are the I in MUDPILES).
- 32M + red urine sometimes when waking in the morning; Dx? → answer = paroxysmal nocturnal hemoglobinuria (PNH).
- Mechanism for PNH? → increased complement-mediated hemolysis caused by deficiency of CD55/59 + deficiency of GPI anchor, which protect RBCs from complement-mediated breakdown.
- Tx for PNH? → eculizumab → monoclonal Ab against complement protein C5.
- Inheritance pattern for hereditary spherocytosis? → answer = autosomal dominant.
- Mechanism for spherocytosis? → answer = deficiency of ankyrin, spectrin, and/or band proteins → results in cytoskeletal disruption and smaller, more spherical RBCs → Qs will sometimes merely have "cytoskeleton" as the answer.
- Notable hematologic parameter in spherocytosis? → answer = ↑ MCHC (mean corpuscular hemoglobin concentration) → only time on USMLE you'll see this variable ↑, however **do not** choose

this for Qs asking you how to Dx spherocytosis. USMLE also wants you to know the spherocytes in hereditary spherocytosis are normochromic, normocytic.

- How to Dx spherocytosis? → answer = osmotic fragility test; if negative answer = eosin-5-maleimide test; the latter is a newer flow cytometry test and is now showing up; osmotic fragility test is the next best step if you're forced to pick between the two, but it can miss up to 25% of cases.
- Tx for hereditary spherocytosis? → answer = splenectomy for those with moderate-severe anemia; the spleen normally clears out the spherocytes, thereby enlarging and also causing chronic anemia.
- 12M + viral infection + spherocytes seen on blood smear + Coombs test positive; Dx? → answer = hemolytic anemia; wrong answer is hereditary spherocytosis; student immediately says, "Omg erratum!" It's not. Chill the fuck out for two seconds. You can get spherocytes in drug-/infection-induced hemolytic anemia → autoantibodies cross-reacting with RBCs (type II hypersensitivity); the key is seeing that the **Coombs test is positive** in drug-/infection-induced spherocytosis because the patient has **antibodies** against RBCs; in contrast, hereditary spherocytosis has zero to do with antibodies; it's a cytoskeletal problem; so of course the Coombs test is negative. Bottom line: yes, you can get spherocytes in things other than spherocytosis, namely drug-/infection-induced AIHA.
- 12M + viral infection + spherocytes seen on blood smear + Coombs test negative; Dx? → answer = hereditary spherocytosis.
- 8M + chronic anemia + father had Hx of splenectomy + smear is shown; inheritance pattern?

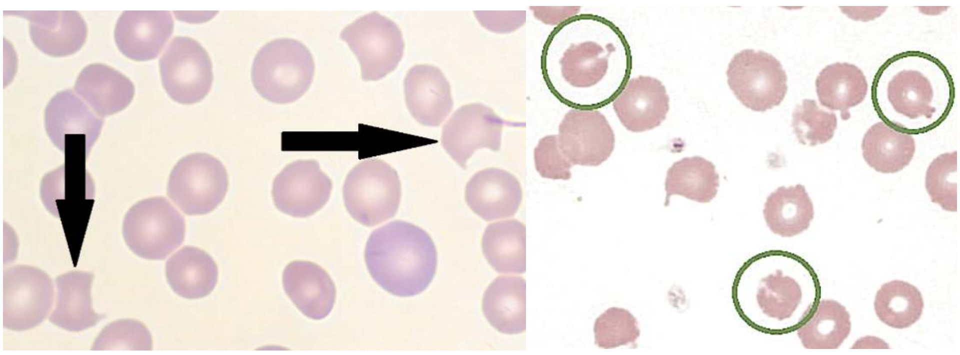


    - Answer = autosomal dominant; smear shows spherocytes. They love this detail about one of the parents having had splenectomy. I've seen students incorrectly answer sickle cell for this Q; sickle cell *causes* autosplenectomy; it isn't treated with splenectomy.
- 40M + recent splenectomy; Q asks, what does the smear show here?

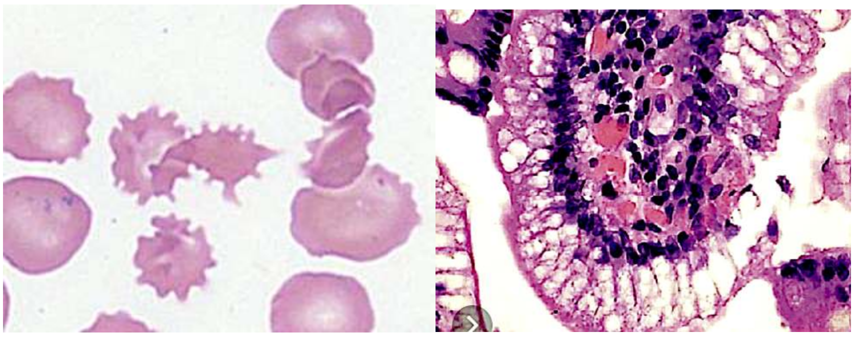


    - Answer = Howell-Jolly bodies within RBCs, which are nuclear remnants; normal finding post-splenectomy.

- Neonate + pathologic jaundice + father had Hx of splenectomy for chronic anemia; Dx? → answer = hereditary spherocytosis.
- 11F + fatigue + Hb 6.5 + MCV 90 + reticulocyte count 9% (NR 0.5-1.5% of RBCs) + mother underwent splenectomy as a youth for "low blood" and recently had a cholecystectomy; what's the most likely mechanism for this patient's condition? → answer = deficiency of erythrocyte spectrin.
- "When is reticulocyte count elevated?" → normal range is 0.5-1.5% of RBCs (according to NBME); if high, indicates hemorrhage, hemolytic anemia, or ↑ RBC turnover (i.e., hereditary spherocytosis; enzyme deficiencies; transfusion reactions; thalassemia when accompanied by severe anemia); if normal or low, then iron deficiency; aplastic anemia; thalassemia not accompanied by severe anemia.
- 18-month-old girl + scleral icterus + pallor + hepatosplenomegaly + Hb 5.6 g/dL + bilirubin 3 mg/dL (normal ~1.0) + smear shows severe hypochromia + nucleated erythrocytes + microcytosis + DNA analysis shows mutation in beta-globin gene; what's the Dx + what arrows do you expect for HbF, HbA2/HbA1 ratio, reticulocyte count; answer = ↑ HbF, ↑ HbA2/HbA1 ratio, ↑ reticulocyte count; Dx is beta-thalassemia accompanied by severe anemia + RBC turnover.
- 12M + treated for infection + smear is shown; Dx?



  - Answer = glucose-6-phosphate dehydrogenase (G6PD) deficiency; left smear shows degmacytes (“bite cells”); right smear shows Heinz bodies.
  - Bite cells are pathognomonic for G6PD deficiency → spleen phagocytoses part, but not all, of RBC, resulting in characteristic “bitten” appearance on smear. Heinz bodies are precipitated / oxidized hemoglobin seen toward the periphery of the RBC.
- Mechanism for G6PD deficiency? → G6PD needed to make NADPH, which acts as a reducing agent (i.e., counteracts oxidation) to protect RBCs → if ↓ NADPH → ↑ oxidation of RBCs → RBCs prone to destruction in setting of stressors such as infection, drugs (i.e., dapsone), or foods (i.e., fava beans).
- Inheritance pattern of G6PD deficiency? → answer = **X-linked recessive (HY).**
- Neonatal girl + pathologic jaundice + hemolytic disease due to enzyme deficiency; Dx? → answer = pyruvate kinase deficiency → second-most common cause of hereditary hemolytic anemia due to an enzyme deficiency (after G6PD deficiency); since G6PD is XR, you know in a girl it can’t be the answer.
- 3F + failure to thrive + blood smear and enterocyte biopsy are both shown; Dx?



  - Answer = abetalipoproteinemia; blood smear shows acanthocytes (spur cells); biopsy of enterocytes shows large, clear fat droplets due to malabsorption (apo-B48 needed for absorption from bowel).
- 82F + found in house unconscious during summer day + body temperature is 105F + blood smear shows acanthocytes; Dx? → answer = liver failure; abetalipoproteinemia is the wrong answer; student says wtf? → heat stroke = end-organ failure secondary to hyperthermia; heat exhaustion = fatigue, but no end-organ failure secondary to hyperthermia; acanthocytes (aka spur cells) can be seen in both liver failure and abetalipoproteinemia; USMLE loves liver failure secondary to heat stroke as a cause of acanthocytosis.

- 2M + SCID + requires blood transfusion for severe anemia; what kind of blood products are most appropriate? → answer = irradiated packed red blood cells.
- What are some ADP P2Y12 receptor blockers? → clopidogrel, prasugrel, ticagrelor, ticlopidine.
- What are some GpIIb/IIIa inhibitors? → abciximab, eptifibatide, tirofiban.
- Drugs that are both anti-platelet agents and vasodilators? → dipyridamole + cilostazol; both mixed cAMP and cGMP phosphodiesterase inhibitors.
- Drugs that are direct-thrombin inhibitors? → bivalirudin, lepirudin, dabigatran, argatroban.
- When are these notably the answer? → once again, for Tx of HIT.
- MOA of fondaparinux? → indirect factor Xa inhibitor (activates antithrombin, causing factor Xa inhibition).
- MOA of apixaban? → direct factor Xa inhibitor.
- Important fibrinolytic? → tPA → used for ischemic stroke within 3-4.5 hours. Streptokinase also fibrinolytic.
- Drugs that are anti-fibrinolytics? → tranexamic acid, aminocaproic acid (can help reverse tPA).
- MOA of methotrexate? → reversible, competitive inhibitor of dihydrofolate reductase; first-line DMARD for RA.
- Side-effects of methotrexate? → pulmonary fibrosis, hepatotoxicity, neutropenia.
- How to mitigate toxicity of methotrexate? → folinic acid (leucovorin rescue), **not** folic acid.
- MOA of 5-fluorouracil (5-FU)? → thymidylate synthase inhibitor.
- MOA of 6-mercaptopurine (6-MP)? → PRPP amidotransferase inhibitor (purine synthesis inhibitor).
- MOA of azathioprine? → metabolized into 6-MP → then inhibits purine synthesis.
- MOA of hydroxyurea? → ribonucleotide reductase inhibitor (pyrimidine synthesis inhibitor).
- MOA of mycophenolate mofetil? → inhibits IMP dehydrogenase (inhibits purine synthesis).
- Microtubule inhibitors? → colchicine, -bendazoles (mebendazole, albendazole), vincristine, vinblastine, taxanes (paclitaxel/docetaxel), griseofulvin.
- Which microtubule inhibitor is the odd one out? → paclitaxel/docetaxel → hyperstabilize microtubules; the others inhibit formation.
- Toxicity of vincristine? → neurotoxic.
- MOA of bleomycin? → causes free radical formation.

- Toxicity of bleomycin? → pulmonary fibrosis.
- MOA of doxorubicin (Adriamycin) / daunorubicin? → DNA intercalators + cause free radicals.
- Toxicity of doxorubicin/daunorubicin? → dilated cardiomyopathy.
- How to mitigate the toxicity of doxorubicin/daunorubicin? → dexrazoxane → chelates free radicals.
- MOA of erlotinib? → EGFR tyrosine kinase inhibitor.
- Notable use of erlotinib? → non-small cell lung cancer (for some reason USMLE cares).
- MOA of cetuximab? → monoclonal Ab against EGFR.
- MOA of imatinib? → bcr/abl tyrosine kinase inhibitor for CML → causes fluid retention (edema).
- MOA of tamoxifen/raloxifene? → selective estrogen-receptor modulators (SERMs).
- Important distinction between tamoxifen and raloxifene? → both are antagonists at ER receptors on breast tissue and are agonists on bone, but only tamoxifen is partial agonist on endometrium (↑ risk of endometrial cancer with tamoxifen).
- MOA of trastuzumab (Herceptin)? → monoclonal Ab against HER-2/neu (ERBB2).
- Toxicity of trastuzumab? → cardiotoxicity.
- MOA of irinotecan/topotecan? → topoisomerase I inhibitors.
- MOA of etoposide/teniposide? → topoisomerase II inhibitors.
- MOA of cyclophosphamide? → guanine N7 alkylating agent.
- Major side-effect of cyclophosphamide? → hemorrhagic cystitis due to metabolite called acrolein.
- How to mitigate toxicity of cyclophosphamide? → Mesna (contains thiol -SH group).
- MOA of tacrolimus? → antagonist at FK506 receptor → decreases intracellular calcineurin → decreases IL-2 transcription → decreases T cell function.
- Important side-effects of tacrolimus → **type II diabetes,** nephrotoxicity.
- MOA of cyclosporin? → antagonist at cyclophilin receptor → decreases intracellular calcineurin → decreases IL-2 transcription → decreases T cell function.
- Important side-effects of cyclosporin → nephrotoxicity, hypertension, gingival hyperplasia, hypertrichosis.
- MOA of sirolimus → antagonist at mTOR → **does not** decrease intracellular calcineurin → decreases *responsiveness* to IL-2 → decreases T cell function.
- Important side-effect of sirolimus → dyslipidemia; notably **not** nephrotoxic.

- Cyclosporin + tacrolimus → ↓ calcineurin + ↓ IL-2 *transcription* + nephrotoxic.
- Sirolimus → no change calcineurin + ↓ *responsiveness* to IL-2 + **not** nephrotoxic.

- Important points about cisplatin → causes oto- and neurotoxicity; all over 2CK-level neuro shelf exams as causing "toxic neuropathy" (chemo-induced neuropathy).
- How to mitigate toxicity of cisplatin? → saline infusion (NaCl, were the Cl^- helps), followed by amifostine.
- MOA of rituximab? → CD20 inhibitor on B cells.

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